Further investigations of irrationality constructions for Dirichlet beta values

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Abstract

This supplement examines attempts to prove irrationality of $\beta(4)$, of at least one value in a small set of even Dirichlet beta values, and of combinations of Catalan's constant with π^2. It gives restrictions on canceling unwanted beta coefficients by polynomial multipliers and determines the real error rates of a Padé family. For a determinant associated with $\beta(2) + \pi^2$, a certified decay estimate exceeds the available denominator cost. The effect of the integer adjustments required to obtain that cost remains uncontrolled. Further results describe the attainable determinant values and explain why several proposed bounds for those adjustments fall short. Exact computations identify additional denominator and cancellation costs for small beta sets. These results do not establish a new irrationality or linear-independence theorem.

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1 The arithmetic question behind the experiments

For $s > 1$, Dirichlet's beta function is

$$\beta(s) = \sum_{j=0}^{\infty} \frac{(-1)^j}{(2j+1)^s}.$$

Its positive odd values are nonzero rational multiples of powers of π. Individual even values are much harder to study. Write $G = \beta(2)$ for Catalan's constant. This supplement concerns attempts to prove irrationality of $\beta(4)$, of a member of a very small set of even beta values, or of combinations $G + \lambda\pi^2$ with rational λ.

The earlier investigation supplement [7] treats the first collection of determinant and integral approaches. The present document covers all 29 subsequent research records from 6–8 November 2026. The finite-count and shifted-interval theorems [8, 9] provide context, but their proofs are not

repeated here. The question now is whether their techniques, or other rational approximations, can reach a specified individual value or a much smaller set.

The intended reader is familiar with partial fractions, prime valuations, basic complex analysis and finite-dimensional linear algebra. The Gram and lattice arguments are introduced through the particular quantity they must control. The first two constructions can be read without the longer mixed-moment analysis.

1.1 What would prove irrationality?

Suppose a fixed finite set of real numbers $\alpha_1, \dots, \alpha_k$ were rational. A common denominator D would make

$$D(a_0 + a_1\alpha_1 + \cdots + a_k\alpha_k) \in \mathbb{Z} \qquad (a_0, \dots, a_k \in \mathbb{Z}).$$

Thus nonzero integer-coefficient forms tending to zero would contradict that assumption. This elementary observation puts three requirements on a construction: the right support, enough smallness after the coefficients are made integral, and nonvanishing along an unbounded sequence.

A small approximation error can conceal the second requirement. For example, $|\beta(4) - p/q|$ must be multiplied by q before it becomes the value of the integer form $q\beta(4) - p$. A spectacularly accurate fraction can still give a large integer form. Several experiments below improve the approximation while falling short at precisely this step.

Definition 1.1. For a nonzero rational vector v, its rational content $\operatorname{cont}_{\mathbb{Q}}(v) > 0$ is the positive rational number for which $v/\operatorname{cont}_{\mathbb{Q}}(v)$ is a primitive integer vector. For a rational polynomial, apply this definition to its coefficient vector. Its primitive normalization divides by this content, with no change of sign.

Lemma 1.2 (Separate the rational constant's cost)**.** *Let $q + \sum_j \rho_j\alpha_j$ have some $\rho_j \neq 0$. Put $g = \operatorname{cont}_{\mathbb{Q}}((\rho_j)_j)$, $A_j = \rho_j/g$, and write $q/g = a/d$ in lowest terms with $d > 0$. The full primitive vector is*

$$(a, dA_1, \dots, dA_k),$$

and its multiplier relative to the original form is d/g.

Proof. The integer vector (A_j) has gcd one, so (dA_j) has gcd d. Since $\gcd(a, d) = 1$, adjoining a leaves gcd one. □

The lemma will explain a concrete failure in Section 7: two expressions have identical beta coefficients but different rational constants. Removing a finite prefix improves the real size, yet the new constant has a larger denominator after normalization by g.

For determinant polynomials the same principle applies to every coefficient, rather than just a pair. If $P_n \in \mathbb{Z}[X]$, $\deg P_n = O(n)$, and $\theta = a/d$ is a fixed rational number, then $d^{\deg P_n} P_n(\theta)$ is integral. On a logarithmic n^2 scale, this fixed-denominator cost is $O(n)$ and is negligible. Allowing d to grow with n destroys that argument. This distinction matters in the rational test evaluations of Section 6.

1.2 How to read the results

The main lines of investigation are summarized below. A limitation of one specified estimate is not a limitation of every estimate for the same construction.

Construction	Established result	Remaining issue
Odd-grid isolation	Fixed polynomial multipliers cannot isolate $\beta(4)$ under the stated degree condition.	Other multipliers, families or nonvanishing mechanisms.
Real Padé forms	Exact real remainders and their exponential rates.	Further common coefficient factors or different evaluation regimes.
Mixed moments	Proved decay is sufficient before integer corrections.	Integer corrections that preserve enough of that decay.
Small beta sets	Complete forms and an exact large-prime residue formula.	A useful arithmetic balance beyond the finite tests.

Sections 2 and 3 can be read independently. Sections 4–6 develop the mixed-moment problem in order: form a matrix whose entries are linear in $G + \pi^2$, bound its determinant, then identify what the integer corrections must accomplish. Section 7 returns to the two- and three-value targets. Section 8 treats other tested variants. The catalog in Appendix A records the shorter investigations and superseded intermediate bounds, with an outcome for each source record.

1.3 Sources, certificates and review scope

The odd-grid construction is inherited from [9], with the rational-function lineage described there. The real Padé construction comes from Kawashima–Poëls [2]; its real interpretation follows the Cauchy-transform viewpoint of Rivoal [6]. The mixed moments and Chebyshev rows come from OpenAI [5]. The small-set derivative construction follows the approach of Krisman [3], including his use of triple zeros for grouped denominator cancellation. Precise attributions are repeated at the relevant proof steps.

The accompanying files give the selected calculations and a complete record catalog. Classical inputs and imported construction lemmas are identified in the text. Rational and interval computations have distinct roles: an exact coefficient identity, a certified finite inequality and a limiting theorem are separate assertions.

The underlying canonical decay estimate and this supplement received separate independent mathematical audits. Appendix B gives their scopes, records the missing hypothesis identified in the correction argument, and describes the accompanying checks.

2 Why polynomial isolation does not transfer cheaply

The shifted-interval construction produces forms involving many even beta values. To isolate $\beta(4)$, one might multiply its rational function by an even polynomial and choose the coefficients to cancel the unwanted values. The obstruction below explains why a fixed polynomial does not provide that cancellation in the single-grid family.

2.1 The single-grid family and its highest coefficient

Fix $m \in \{1, 3\}$, $r \geq 1$, and an even integer $a > m(2r + 1)$. For positive odd n, put

$$P_n(t) = \prod_{j=0}^{(2r+1)n-1} (t - 2rn + 2j + 1), \qquad R_n(t) = \frac{\kappa_n P_n(t)^m}{\prod_{h=0}^{n} (t + 2h)^a}.$$

Here $\kappa_n > 0$ is rational. The normalizations used for prime dilation will be specified below; they do not affect coefficient cancellation. Let $k = m - 1$ and take a nonzero homogeneous multiplier

$$W_n(t) = n^E w((t+n)/n), \qquad w(x) = \sum_j w_j x^{2j}, \qquad \deg w \le E.$$

For all sufficiently large odd n, the associated linear form is the convergent alternating sum

$$S_n(W) = \frac{1}{k!} \sum_{v=0}^{\infty} (-1)^v \, \frac{\mathrm{d}^k}{\mathrm{d}t^k} (W_n(t) R_n(t)) \Big|_{t=2v+1}.$$

Write $W_n R_n = \sum_{h,i} c_{i,h} (t+2h)^{-i}$. Each pole contributes a shifted beta series; its finite initial segment contributes to a rational constant q_n. Thus

$$S_n(W) = q_n + \sum_{s=m+1, m+3, \ldots, a+m-1} \rho_s(n) \beta(s),$$

where differentiation and the shift by h give

$$\rho_{i+k}(n) = \binom{i+k-1}{k} \sum_{h=0}^{n} (-1)^h c_{i,h}. \tag{2.1}$$

The odd coefficients vanish by reflection. The possible even indices run from $m+1$ through $a+m-1$.

The highest coefficient is especially useful because no local derivatives are needed to calculate it. Set

$$g_{n,h} = \frac{|P_n(-2h)|^m}{(h!(n-h)!)^a}, \qquad x_{n,h} = 1 - 2h/n.$$

Counting the negative numerator factors gives

$$\rho_{a+k}(n) = \binom{a+k-1}{k} \kappa_n 2^{-an} (-1)^{rn} n^E \sum_{h=0}^{n} g_{n,h} w(x_{n,h}). \tag{2.2}$$

The character signs have disappeared from the last sum: its weights are positive. Their concentration near $h = n/2$ prevents a fixed nonzero polynomial from continuing to cancel them.

Lemma 2.1. *For fixed allowed parameters and any nonzero even real polynomial w, $\sum_h g_{n,h} w(x_{n,h})$ is nonzero for every sufficiently large odd n. Its eventual sign is the sign of the first nonzero coefficient of w.*

Proof. Splitting the numerator into its negative and positive factors gives

$$|P_n(-2h)| = \frac{2^{(2r+1)n}}{\pi} \Gamma(rn + h + \tfrac{1}{2}) \Gamma((r+1)n - h + \tfrac{1}{2}).$$

Uniform Stirling bounds, including $h = 0, n$, show that

$$\log g_{n,h} = C_n + n\Phi(h/n) + O(\log(n+2)),$$

where C_n is independent of h and

$$\Phi(t) = m\{(r+t)\log(r+t) + (r+1-t)\log(r+1-t)\} - a\{t \log t + (1-t)\log(1-t)\}.$$

Use $0 \log 0 = 0$. Its second derivative on $(0,1)$ is

$$\Phi''(t) = m\left(\frac{1}{r+t} + \frac{1}{r+1-t}\right) - a\left(\frac{1}{t} + \frac{1}{1-t}\right) < 0.$$

Symmetry therefore puts the unique maximum at $1/2$. For every fixed $\epsilon > 0$, the weight outside $|x_{n,h}| < \epsilon$ is at most $n^C e^{-c_\epsilon n}$ times a central weight, after summing over h.

Write $w(x) = cx^{2j} + O(x^{2j+2})$, $c \neq 0$, and choose ϵ so that w has sign c for $0 < |x| < \epsilon$ and $|w(x)| \geq |c||x|^{2j}/2$ there. Odd n ensures $x_{n,h} \neq 0$. At $h = (n-1)/2$, $|x_{n,h}| = 1/n$, so this one central term is at least $|c|n^{-2j} g_{n,h}/2$ in absolute value. The outer terms are exponentially smaller. All other central terms have the same sign. □

2.2 The exceptional case where the highest value is the target

Usually $a + m - 1 > 4$, so Lemma 2.1 already leaves an unwanted value. The only exception is $m = r = 1$, $a = 4$. If $E < a$, the possible even homogeneous multipliers are constants, multiples of n, or $c_0 n^2 + c_1(t+n)^2$ and their multiples by n. The extra power of n is irrelevant. We must show that ρ_2 survives.

Put $X_h = n - 2h$ and let $\mathbb{E}_n$ denote averaging with weights $g_{n,h}$. At the pole $-2h$, let ℓ_1, ℓ_2 be the first two derivatives of the logarithm of the regular part of R_n. Direct logarithmic differentiation gives

$$\ell_1(h) = O_{2n-h} - O_{n+h} - 2(H_{n-h} - H_h),$$
$$\ell_2(h) = -O^{(2)}_{2n-h} - O^{(2)}_{n+h} + H^{(2)}_{n-h} + H^{(2)}_h,$$

where $H_j = \sum_{v=1}^{j} v^{-1}$ and $O_j = \sum_{v=0}^{j-1} (2v+1)^{-1}$; superscript (2) squares the denominators. On $n/4 \leq h \leq 3n/4$,

$$\ell_1 = O(|X_h|/n + 1/n), \qquad \ell_2 = \pi^2/12 + O(1/n).$$

The second constant is the difference between the two-sided even and odd square sums. Globally $\ell_1 = O(\log n)$ and $\ell_2 = O(1)$.

Here $\Phi''(1/2) = -44/3$. A Taylor expansion at the strict maximum, with $h = n/2 + y\sqrt{n}$, gives the Gaussian weight $\exp(-22y^2/3)$. Uniform Stirling bounds on a central interval and exponential suppression outside it justify the Riemann sums, also after multiplication by y^2 or y^4. Consequently

$$\mathbb{E}_n X_h^2 \sim 3n/11, \qquad \mathbb{E}_n X_h^4 = O(n^2).$$

It follows that

$$\mathbb{E}_n[(\ell_1^2 + \ell_2)/2] = \pi^2/24 + O(1/n),$$
$$\mathbb{E}_n[X_h^2(\ell_1^2 + \ell_2)/2 + 2X_h\ell_1 + 1] = \pi^2 n/88 + o(n).$$

Apart from a common nonzero factor, ρ_2 is $c_0 n^2$ times the first expression plus c_1 times the second. If $c_0 \neq 0$, its quadratic term survives; otherwise its linear term survives.

Theorem 2.2. *In the single-grid family above, a fixed nonzero homogeneous even multiplier of degree $E < a$ cannot give forms supported only on $1, \beta(4)$ for infinitely many odd indices. For every sufficiently large odd index an unwanted beta coefficient is nonzero.*

2.3 Polynomial dependence on the index

A polynomial $W(n,t)$ may have several homogeneous degrees. Write $W = \sum_{e \geq e_0} W_e$ with $W_{e_0} \neq 0$, where W is even about $t = -n$. The prime-dilation argument transfers the lowest degree to a fixed seed. For this purpose take

$$\kappa_n = 2^{an}(n!)^{a-m(2r+1)}.$$

This is the single-grid normalization of [9]; a fixed integer factor raised to n would make no difference to the following congruence away from its prime divisors.

Lemma 2.3. *Fix an odd seed $\ell \geq 1$. If $e_0 < a$, then for every sufficiently large prime p and $1 \leq i \leq a$,*

$$p^{a-e_0-i}\rho_{i+k}^{W}(\ell p) \equiv \rho_{i+k}^{W_{e_0}}(\ell) \pmod{p}.$$

The scaled quantities are p-integral, even when the displayed exponent is negative.

Proof. In each numerator block the factors divisible by p, after division by p, reproduce the seed block. The unselected factors have unit $(-1)^{\ell(2r+1)}$. Also $(\ell p)!/p^{\ell} \equiv (-1)^{\ell}\ell! \pmod{p}$ for $p > \ell$. Raising these units to the exponents in R_n makes their signs cancel, since a is even. The selected denominator poles are $h = py$, $0 \leq y \leq \ell$, and the other denominator unit is one. Counting powers gives $p^a R_{\ell p}(pt) \equiv R_{\ell}(t)$. Moreover $p^{-e_0}W(\ell p, pt) \equiv W_{e_0}(\ell, t)$.

At a selected pole, remove the pole factor. Every remaining denominator has unit constant term in the local power series, so reduction modulo p commutes with extraction of its Taylor coefficients. Thus $p^{a-e_0-i}c_{i,py}^{W}(\ell p) \equiv c_{i,y}^{W_{e_0}}(\ell)$. At a nonselected pole, the normalization and numerator give valuation at least $\ell a - u$ in Taylor degree u. The other poles in its class cost at most $a(\ell-1)+v$ in inverse-series degree v. In total degree $a-i$, the resulting coefficient has valuation at least i. Its scaled contribution therefore has valuation at least $a - e_0 > 0$. Summing over the selected poles, using $(-1)^{py} = (-1)^{y}$, proves the claim with the differentiation factor in (2.1). □

Theorem 2.4. *Under the same single-grid assumptions, a nonzero polynomial multiplier $W(n,t) \in \mathbb{Q}[n,t]$, even about $-n$, whose least homogeneous degree is less than a, cannot isolate $\beta(4)$ at every sufficiently large odd index.*

Proof. If all unwanted coefficients vanished on that tail, fix any odd seed ℓ and apply Lemma 2.3 along $n = \ell p$. Each unwanted seed coefficient for W_{e_0} is zero modulo every sufficiently large prime, hence is zero as a rational number. This holds for every seed and contradicts Theorem 2.2. □

The least-degree condition is the one preserving strict valuation separation in the existing prime-dilation nonvanishing argument. The theorem does not address multipliers outside that condition, nonpolynomial dependence on n, or arbitrary sparse subsequences. For multiple grids, fixed grid weights alone also have a basic limit: a generalized Vandermonde matrix at distinct positive nodes has full rank. There are more unwanted columns than available grid weights in the existing construction. This excludes a universal projection, not cancellations after some actual coefficients have separately vanished.

2.4 Adaptive cancellation and telescoping kernels

Index-dependent multipliers do isolate $\beta(4)$ in all 27 exact tests recorded in [R25]. After complete primitive reduction, every one of the 18 forms at $n = 3, 5$ has absolute value greater than one. This is finite evidence about their arithmetic, not an extension of the theorem to every adaptive multiplier.

An exact cancellation kernel can also be misleading because it may cancel the whole form. In the smallest case $m = r = 1, a = 4$, put $x = t + n$. For any polynomial $F(n, y)$ odd in y, define

$$W_F(n,x) = \tfrac{1}{2}\{(x-3n-1)(x+n)^4F(n,x-1) \\ + (x+3n+1)(x-n)^4F(n,x+1)\}.$$

The exact ratio

$$\frac{R_n(t+2)}{R_n(t)} = \frac{x+3n+1}{x-3n+1}\left(\frac{x-n}{x+n+2}\right)^4$$

shows that $W_F R_n = J_n(t) + J_n(t+2)$, with $J_n(t) = \frac{1}{2}(x-3n-1)(x+n)^4F(n,x-1)R_n(t)$. The alternating sum at positive odd integers telescopes to $J_n(1) = 0$ for all sufficiently large odd n. Its beta coefficients cancel separately as well, by shifting the principal parts.

For $F = y$, the multiplier is

$$x^6 + (1 - 5n - 6n^2)x^4 + (6n^2 + 10n^3 - 11n^4)x^2 + n^4 + 3n^5.$$

The map $F \mapsto W_F$ has leading x coefficient one and increases the x degree by five. Successively subtracting these telescopers therefore gives a unique representative

$$A(n) + B(n)x^2 + C(n)x^4$$

for every even polynomial multiplier modulo the telescoping subspace. Finite coefficient matrices through total degree 16 have exactly the telescoping kernels described in the catalog. They do not prove independence of the three remaining sequences over $\mathbb{Q}(n)$.

3 Real remainders from a simultaneous Padé construction

The preceding construction begins with many beta values and tries to remove some of them. The present family has a remainder involving $\beta(4)$ alone from the start. It comes from the $m = 5$ construction of Kawashima–Poëls [2], which approximates two formal functions with a common denominator polynomial. The two remainders are separate; simultaneous approximation does not introduce a $\beta(2)$ term into the weight-four remainder.

The first task is to give that formal remainder an actual real meaning. The second is to estimate it when both the degree and evaluation point grow. Formal order of approximation at a fixed degree does not answer the latter question.

3.1 Functions and polynomials

For $s = 2, 4$ and $\Re z > 0$, set

$$F_s(z) = (s-1)!\sum_{j\ge 0}\frac{(-1)^j}{(z+j)^s} = \int_0^\infty \frac{u^{s-1}e^{-zu}}{1+e^{-u}}\,\mathrm{d}u.$$

The shift identity $F_s(z)+F_s(z+1) = (s-1)!z^{-s}$ determines its formal asymptotic Laurent coefficients $c_{s,j}$ by

$$2c_{s,j} = (s-1)!\,[j = s-1] - \sum_{v<j}(-1)^{j-v}\binom{j}{v}c_{s,v}. \tag{3.1}$$

The indexing convention is $F_s(z) \sim \sum_{j\ge 0} c_{s,j}z^{-j-1}$; call the corresponding formal series Θ_s. In (3.1), $[j = s-1]$ is one when $j = s-1$ and zero otherwise. A finite Taylor expansion of the integrand

near zero, with a separate exponentially small tail, proves that these are the asymptotic coefficients of F_s. No convergence of the Laurent series at a finite real point is required.

Write $(x)_k = x(x+1)\cdots(x+k-1)$, $T_k(z) = (z-k/2)_k$, and

$$\mathcal{R}H(z) = \frac{H(z+1/2) + H(z-1/2)}{2z-1}.$$

For positive even n and $\ell = 0, 1, 2$, let

$$H_{n,\ell}(z) = \frac{T_n(z)^5 T_{2\ell}(z)}{(n!)^4}, \qquad P_{n,\ell} = \mathcal{R}^n H_{n,\ell}.$$

The exact divisibility in these operations, reflection $P_{n,\ell}(1-z) = P_{n,\ell}(z)$, and degree $4n+2\ell$ are the specialization of [2, Lemma 6.5]. Define a rational functional by $\varphi_s(t^j) = c_{s,j}$ and put

$$Q_{n,s,\ell}(z) = \varphi_s\left(\frac{P_{n,\ell}(z) - P_{n,\ell}(t)}{z-t}\right).$$

This is the polynomial part of the formal product $P_{n,\ell}\Theta_s$. The source's approximation theorem gives a remainder of order $O(z^{-2n-2})$, with its constant depending on n.

Two other source inputs will be used explicitly. Its integer-basis theorem [2, Theorem 8.1] writes $P_{n,\ell}$ as an integer combination of $(z)_j(1-z)_j/(j!)^2$, $j \le 2n+2$. Its rank theorem [2, Theorem 7.1] says that the three columns

$$(P_{n,\ell}, Q_{n,2,\ell}, Q_{n,4,\ell})^T, \qquad \ell = 0, 1, 2,$$

have a nonzero rational constant determinant. These are polynomial statements; they do not require a p-adic evaluation.

3.2 From a formal remainder to a real integral

Put $\mu(y) = (2\cosh \pi y)^{-1}$ and $t_y = 1/2 + iy$. The Fourier identity

$$\int_{\mathbb{R}} \mu(y) e^{iuy}\, \mathrm{d}y = (2\cosh(u/2))^{-1}$$

can be proved by a rectangular contour of height i; it is also the Euler-moment identity used in Rivoal's real construction [6]. Fubini's theorem gives, for $\Re z > 1/2$,

$$F_s(z) = (s-1)! \int_{\mathbb{R}} \frac{\mu(y)}{(z-t_y)^s}\, \mathrm{d}y, \qquad \varphi_s(A) = \int_{\mathbb{R}} \mu(y) A^{(s-1)}(t_y)\, \mathrm{d}y.$$

For the first interchange an absolute majorant is $u^{s-1}e^{-(\Re z - 1/2)u}\mu(y)$. For the second identity compare exponential moment generating functions and then coefficients. Subtracting the polynomial part yields the exact remainder

$$\begin{aligned} E_{n,s,\ell}(z) &= P_{n,\ell}(z)F_s(z) - Q_{n,s,\ell}(z) \\ &= i^{s-1} \int_{\mathbb{R}} \mu^{(s-1)}(y) \frac{P_{n,\ell}(t_y)}{z-t_y}\, \mathrm{d}y. \end{aligned} \tag{3.2}$$

Integration by parts proves the last expression; exponential decay of the weight kills every boundary term.

For an integer $N \geq 1$, define $S_N(4) = \sum_{j<N} (-1)^j/(2j+1)^4$. Splitting the defining beta series at N gives

$$F_4(N + \tfrac{1}{2}) = 96(-1)^N \{\beta(4) - S_N(4)\}.$$

Consequently $E_{n,4,\ell}(N + 1/2) = b + a\beta(4)$, where

$$a = 96(-1)^N P_{n,\ell}(N + \tfrac{1}{2}), \qquad b = -aS_N(4) - Q_{n,4,\ell}(N + \tfrac{1}{2}). \tag{3.3}$$

The rational tail in b is part of the construction and must be cleared. The rank theorem above ensures that at least one of these three forms is nonzero at every such n, N: otherwise the last row of the constant determinant would be $F_4(N + 1/2)$ times the first.

For example, at $n = 2, \ell = 0, N = 1$,

$$P(3/2) = 31681/4096, \quad Q_4(3/2) = 2223/256, \quad E(3/2) = -\frac{39}{256}\{4874\beta(4) - 4817\}.$$

The primitive integer form has absolute value about 3.12. The small prefactor in the displayed raw form provides no arithmetic saving.

3.3 Transfer the difference operators to the kernel

Estimating the large polynomial P term by term loses its cancellations. Moving its difference operators to the Cauchy kernel gives products with a much clearer asymptotic balance. Define

$$A_n(y) = \prod_{j=1}^{n/2} \{y^2 + (j - \tfrac{1}{2})^2\},$$
$$B_\ell(y) = \prod_{j=1}^{\ell} \{y^2 + (j - \tfrac{1}{2})^2\},$$
$$D_{n,N}(y) = \prod_{j=0}^{n} \{(N - n/2 + j)^2 + y^2\}.$$

An empty product is one.

Proposition 3.1. *For real $N \geq n/2$,*

$$E_{n,4,\ell}(N + \tfrac{1}{2}) = \frac{2(-1)^\ell}{(n!)^3} \int_0^\infty \frac{y\mu'''(y) A_n(y)^5 B_\ell(y)}{D_{n,N}(y)} \, \mathrm{d}y.$$

At $N = n/2$ the integrand is interpreted continuously at $y = 0$.

Proof. On the upward line $\Re t = 1/2$, let $w_j(t)$ be the third derivative of $\csc \pi(t + j/2)$. Oddness of w_0 about $1/2$ allows replacement of the Cauchy kernel by its odd part. With

$$K_j(t, z) = \frac{j!(2t-1)}{(z - t - j/2)_{j+1} (z + t - 1 - j/2)_{j+1}},$$

equation (3.2) becomes $E(z) = -(4i)^{-1} \int w_0(t) P(t) K_0(t, z) \, \mathrm{d}t$. Moving one difference operator from P replaces K_j by

$$\frac{K_j(t + 1/2, z)}{2t} - \frac{K_j(t - 1/2, z)}{2t - 2} = K_{j+1}(t, z).$$

For a direct check, put $a = z - t - (j+1)/2$ and $b = z + t - (j+3)/2$. The numerator after taking a common denominator is $j!\{(a+j+1)b - a(b+j+1)\} = (j+1)!(2t-1)$.

After j transfers the remaining polynomial is divisible by T_j^5 by the source divisibility lemma. In the next transfer the vertical lines shift to real parts zero and one. The poles of w_{j+1} in the crossed strips have order at most four and are canceled by zeros of T_{j+1}^5. The apparent factors $2t$ and $2t-2$ have canceled in the kernels. For $N > n/2$ the remaining kernel poles lie outside the respective crossed strips. Exponential decay kills the horizontal closing segments, so no residue is lost.

After n transfers, put $t = 1/2 + iy$. Then

$$w_n = (-1)^{n/2} 2i\mu'''(y), \quad T_n = (-1)^{n/2} A_n(y), \quad T_{2\ell} = (-1)^\ell B_\ell(y), \quad K_n = 2iyn!/D_{n,N}(y).$$

Substitution and pairing y with $-y$ give the claimed normalization and sign. As $N \downarrow n/2$, $\mu'''(y) = O(y)$ cancels the only new zero denominator: $y^2/((N-n/2)^2 + y^2) \le 1$. The other factors supply an integrable majorant near zero and infinity. Dominated convergence gives the endpoint. □

3.4 The actual exponential rate

For $r \ge 1/2$ define

$$\Phi_r(u) = 3 - \pi u + 5\int_0^{1/2} \log(u^2 + x^2)\,\mathrm{d}x - \int_{r-1/2}^{r+1/2} \log(u^2 + x^2)\,\mathrm{d}x.$$

The constant 3 comes from $(n!)^{-3}$, the term $-\pi u$ from the hyperbolic weight, and the two integrals from the numerator and denominator products after setting $y = nu$. All powers of n cancel.

Theorem 3.2. *The function Φ_r has a unique maximum at $u_r > 1/2$. Put $\Lambda(r) = \Phi_r(u_r)$. If n is even, $N \ge n/2$ is integral, and $N/n \to r \ge 1/2$, then for each $\ell = 0, 1, 2$,*

$$\lim \frac{1}{n} \log |E_{n,4,\ell}(N + \tfrac{1}{2})| = \Lambda(r).$$

Its sign is eventually $(-1)^{\ell+1}$.

Proof. For $u > 0$,

$$\Phi_r'(u) = -\pi + 10\arctan(1/(2u)) - 2\arctan((r+1/2)/u) + 2\arctan((r-1/2)/u),$$
$$\Phi_r''(u) = -\frac{5}{u^2 + 1/4} + \frac{2(u^2 - ab)}{(u^2 + a^2)(u^2 + b^2)} \le -\frac{3}{u^2 + 1/4} < 0,$$

where $a = r - 1/2$, $b = r + 1/2$. The derivative is positive at $1/2$ and tends to $-\pi$ at infinity. This proves the unique maximum.

The sign issue in the integral is localized:

$$\mu'''(y) = \frac{\pi^3}{2} \operatorname{sech}(\pi y) \tanh(\pi y)\{6\operatorname{sech}^2(\pi y) - 1\}.$$

It is positive up to the fixed value $y_0 = \operatorname{arcosh}(\sqrt{6})/\pi$ and negative afterwards. All the product factors are positive. The saddle region $y \sim nu_r$ therefore has one sign and is separated from the bounded region where the sign is opposite.

Put $r_n = N/n$. On compact positive u intervals, product Riemann sums and Stirling's formula give $n\Phi_{r_n}(u) + O(\log n)$, uniformly for r_n in a fixed compact subset of $[1/2, \infty)$. The product estimates retain this logarithmic error down to $u = y_0/n$ on bounded intervals: bracket each

monotone Riemann sum by its endpoint sums, whose difference is $O(\log n)$. The integral formula for $\Phi_r(u)$ is jointly continuous for $r \geq 1/2$ and $u \geq 0$. Hence $r_n \to r$ gives $\Phi_{r_n} \to \Phi_r$ uniformly on bounded u intervals. After division by n, the limiting profile is therefore Φ_r; no rate of convergence of r_n is needed.

To bound $0 \leq y \leq y_0$ separately, use $A_n(y) \leq CA_n(0)$, which follows from $\log(1+x) \leq x$ and the convergent sum of $(j-1/2)^{-2}$. Put $k = N - n/2$. For $k \geq 1$ this part is at most

$$\frac{CA_n(0)^5}{(n!)^3(k)_{n+1}^2},$$

and for $k = 0$, after canceling y^2, at most $CA_n(0)^5/(n!)^5$. The gamma identities for these products give upper rate $\Phi_r(0)$ uniformly even when $k/n \to 0$. This is strictly below $\Phi_r(u_r)$.

For the infinite tail, use $A_n(nu) \leq n^n(u^2+1/4)^{n/2}$, $D_{n,N}(nu) \geq (nu)^{2n+2}$ and $n! \geq (n/e)^n$. Apart from polynomial factors, the tail is bounded by

$$\exp\{n[3 - \pi u + \tfrac{5}{2}\log(u^2+1/4) - 2\log u]\}.$$

For sufficiently large u the bracket is at most $-\pi u/2$. Laplace's estimate near the strict maximum now supplies the claimed rate. The opposite-sign part has smaller rate, so cannot cancel it. □

The saddle and rate admit exact rational certification. If $w_r = u_r^2$, then w_r is the unique root greater than $1/4$ of

$$64w^3 + (64r^2 - 16)w^2 - (160r^2 + 20)w + 20r^2 - 3 = 0.$$

This follows by taking the real part of $(u+i/2)^5(u+i(r-1/2))(u-i(r+1/2))$ in the saddle equation. Descartes' rule and the signs at zero and $1/4$ select the larger positive root. The arctangent terms cancel at the saddle, giving

$$\begin{aligned}\Lambda(r) = {} & \tfrac{5}{2}\log(w_r + 1/4) - (r+1/2)\log(w_r + (r+1/2)^2) \\ & + (r-1/2)\log(w_r + (r-1/2)^2).\end{aligned}$$

For example $w_2 = 7/4$ and $\Lambda(2) = -2\log 2$ exactly.

3.5 Intersect two denominator bounds

Let $d_j = \operatorname{lcm}(1, \ldots, j)$ and let a superscript "odd" mean the odd part. Put

$$J = 2n+2, \quad R_n^{\mathrm{ar}} = \gcd(n!, d_J)^{\mathrm{odd}}, \quad L_{n,N} = \operatorname{lcm}(R_n^{\mathrm{ar}}, d_{2N-1}^{\mathrm{odd}}).$$

Proposition 3.3. *Both coefficients in (3.3) become integral after multiplication by*

$$M_{n,\ell,N} = 2^{4n+2\ell-4+4v_2(n!)} L_{n,N}^4.$$

For $N/n \to r \geq 1/2$, its logarithmic rate is $8\log 2 + 8r$.

Proof. First $(n!)^4 P_{n,\ell} \in \mathbb{Z}[z]$. Expand the monic integer seed $T_n^5 T_{2\ell}$ in the monic basis T_{n+2j} by successive leading terms. Reflection preserves the even degrees and divisibility by T_n forces the final remainder to vanish. The identity $\mathcal{R}T_k = T_{k-1}$ proves the assertion. Recurrence (3.1) gives $c_{4,0} = c_{4,1} = c_{4,2} = 0$ and $2^{j-3}c_{4,j} \in \mathbb{Z}$ for $j \geq 3$. Thus the odd denominator of $Q_{n,4,\ell}(N+1/2)$ divides the odd part of $(n!)^4$.

The factorial-basis calculation of [2, Sections 5 and 9] gives a second bound. To state the needed arithmetic explicitly, its coefficients satisfy, for $j \geq 1$,

$$2b_j = \sum_{u+v=j-1} \frac{u!^2 v!^2}{(u+1)(2u+1)!(2v+1)!}, \qquad \Theta_4(z) = \sum_{j\geq 1} \frac{b_j (2j+1)!}{(z)_{j+1}(1-z)_{j+1}}.$$

One has $2d_j^2(2j+1)!b_j/(j!)^2 \in \mathbb{Z}$: a summand multiplied by $(2j+1)!/(j!)^2$ is $2\binom{2j+1}{2u+2}(u!v!/j!)^2$, and $j!/(u!v!)$ divides d_j. At each prime power its valuation contribution is zero or one because $u+v=j-1$. Likewise $(j+1)\binom{k}{j+1}$ divides d_k. Canceling the factorials against an integer basis term of P shows that the odd denominator of $Q(N+1/2)$ divides $(d_J^{\mathrm{odd}})^4$. Half-integer factorial evaluations introduce no other odd primes; indeed $2^{2k}(x)_k/k! \in \mathbb{Z}$ for half-integral x. Intersecting the two bounds gives $(R_n^{\mathrm{ar}})^4$.

The polynomial P has no odd denominator at a half-integer, by its integer factorial basis. The finite tail $S_N(4)$ therefore adds only $(d_{2N-1}^{\mathrm{odd}})^4$. Taking their lcm gives $L_{n,N}^4$.

For powers of two let $D = 4n + 2\ell$. A term $P_d c_{4,j} z^{d-j-1}$ of the polynomial part has valuation at least $-4v_2(n!) - (j-3) - (d-j-1) \geq -4v_2(n!) - D + 4$. The coefficient of $\beta(4)$ includes 96, with valuation five; $S_N(4)$ has odd denominator. The stated power of two clears both.

Finally $d_n^{\mathrm{odd}} \mid R_n^{\mathrm{ar}}$, and any additional prime power over d_n has base at most $\sqrt{J}$. Its total logarithmic cost is $O(\sqrt{n}\log n)$. Hence $\log L_{n,N} = \log d^{\mathrm{odd}}_{\max(n,2N-1)} + O(\sqrt{n}\log n)$. The prime number theorem and $v_2(n!) = n + O(\log n)$ prove the rate. □

3.6 The gap that remains

Combine the two proved rates and write $\mathcal{G}(r) = 8\log 2 + 8r + \Lambda(r)$. This is the logarithmic rate of the cleared forms; a negative value would make them tend to zero.

r	Remainder rate	Clearing cost	Sum $\mathcal{G}(r)$
1/2	−0.21025504	9.54517744	9.33492240
1	−0.61293790	13.54517744	12.93223954
2	−1.38629436	21.54517744	20.15888308

This is an asymptotic comparison for the explicit multiplier, not a fit to finite values. Differentiation at the unique maximum gives

$$\Lambda'(r) = \log \frac{u_r^2 + (r-1/2)^2}{u_r^2 + (r+1/2)^2}.$$

Its negative is $\int_{r-1/2}^{r+1/2} 2x/(u_r^2+x^2)\,\mathrm{d}x \leq 1/u_r < 2$. Therefore $\mathcal{G}'(r) > 6$: moving the evaluation point farther right makes this balance worse. The minimum is $\mathcal{G}(1/2) = 9.33492239999\ldots$.

This does not lower-bound the primitive forms. If the two cleared coefficients have gcd g_n, their primitive value is $|ME|/g_n$. Decay would require $\liminf n^{-1}\log g_n \geq \mathcal{G}(r)$. No upper bound excluding that saving has been proved. Evaluation below $N = n/2$, combinations of the three forms and modified seeds are also outside this rate theorem. The finite tests and their precise ranges are recorded in [R26]–[R27].

4 Mixed moments for Catalan's constant and π^2

4.1 Keep the two periods in a prescribed ratio

A rational relation between 1, $G = \beta(2)$ and π^2, with a nonzero coefficient of G, would make $G + \lambda\pi^2$ rational for some rational λ. This suggests retaining the two periods in a fixed ratio inside each

matrix entry, before taking a determinant. Since π^2 is irrational, proving that $G + \lambda\pi^2$ is irrational for every rational λ would establish independence of $1, G, \pi^2$. The detailed decay calculation below treats the single test case $\lambda = 1$. The moments of OpenAI [5, Section 2] make that operation particularly direct. The adaptation below is from [R11].

Put $c_j = 4^{-j}\binom{2j}{j}$ for nonnegative integers j, and $c_j = 0$ otherwise. For a polynomial $P(t) = \sum P_i t^i$, define

$$A_j(P) = \sum_i P_i c_{(j-i-1)/2}, \qquad B_j(P) = \sum_i P_i c_{(i-j)/2}.$$

Thus $A_j(P)$ is the coefficient of t^j in $tP(t)/\sqrt{1-t^2}$ at zero, whereas $B_j(P)$ is the coefficient of t^j in the polynomial part at infinity

$$\Omega(P) = \left[\frac{P(t)}{\sqrt{1-t^{-2}}}\right]_+ .$$

The square roots are expanded with constant term one. Define the moments

$$M(P,j) = \int_{-1}^{1}\int_0^1 \frac{|t|P(t)s^j}{\sqrt{1-t^2}(1-ts)}\,\mathrm{d}s\,\mathrm{d}t,$$
$$Z(D,j) = \int_0^1\int_0^1 \frac{D(t)s^j}{1-ts}\,\mathrm{d}s\,\mathrm{d}t$$

and, for $Q(s) = \sum Q_j s^j$, write $M(P,Q) = \sum Q_j M(P,j)$ and $Z(D,Q) = \sum Q_j Z(D,j)$. Their period coefficients are

$$M(P,j) = M^0(P,j) + 4B_j(P)G + \tfrac{3}{2}A_j(P)\zeta(2), \qquad Z(D,j) = Z^0(D,j) + D_j\zeta(2), \tag{4.1}$$

where the superscript 0 denotes the rational part. These are the moment identities in [5, (17)].

Suppose D_0 has coefficient $A_j(P)$ in every column being used, and $E(P)$ has coefficient $B_j(P)$ there. Replace D_0 by

$$D_\lambda = D_0 - 16\lambda E(P).$$

Then $M(P,j) - \frac{3}{2}Z(D_\lambda, j)$ has the form

$$\text{rational} + 4B_j(P)(G + \lambda\pi^2). \tag{4.2}$$

The factor 16 supplies $24\lambda B_j\zeta(2) = 4\lambda B_j\pi^2$. For example, $P = t^2$ gives $\Omega(P) = t^2 + \frac{1}{2}$. In columns $0 \le j < 4$, take $D_0 = t^3$ and $E = t^2 + \frac{1}{2}$. The new polynomial is $t^3 - 16\lambda t^2 - 8\lambda$, and the coefficients of $G + \lambda\pi^2$ are $2, 0, 4, 0$.

4.2 Rows, columns and the canonical matrix

We now assemble n matched rows and n column polynomials into a square matrix. Each entry will be rational plus a rational multiple of $\theta_\lambda = G + \lambda\pi^2$. Replacing θ_λ by a variable X makes its determinant a rational polynomial of degree at most n. Our goal is to make the value at θ_λ small enough to survive clearing the polynomial's denominators.

The columns use powers s^i in a band $b \le i < L$. Matching the period coefficients in this band fixes the target; the b lower powers remain available for arithmetic corrections. Zeros of orders h at $t = 1$ and q at $s = 1$ improve the real estimates. A further parameter g counts pairs of rows that will supply fixed rational rows.

The cutoffs are related by $L = n + b + q$, $C = L + g$ and $H = C + h$. Here L is the first unused column degree, $C - 1$ is the common power of t in the row formulas, and $H - 1$ bounds the row degrees. We use the following fixed proportions throughout Sections 4–6:

$$\begin{aligned} n &= 48N, \quad b = 7N, \quad q = g = 4N, \quad h = 2N, \\ L &= 59N, \quad C = 63N, \quad H = 65N. \end{aligned} \tag{4.3}$$

Only N tends to infinity. This choice leaves room for the corrections while balancing the denominator cost against endpoint vanishing; the comparison of the two rates will be made in Section 5. The rows and columns inherited from [5] are

$$\begin{aligned} P_r(t) &= (1-t)^h t^{C-1} T_{|r-g|}(1/t), \\ D_r(t) &= \operatorname{sgn}(r-g)(1-t)^h t^{C-1} U_{|r-g|-1}(1/t), \\ Q_j(s) &= s^{b+j}(1-s)^q \qquad (0 \le r, j < n), \end{aligned} \tag{4.4}$$

with $D_g = 0$. Here T_d, U_d are Chebyshev polynomials. The contact identity of [5, (5)] is

$$tP_r(t)/\sqrt{1-t^2} - D_r(t) = O(t^L).$$

Since the column polynomials use only the raw powers s^i with $b \le i < L$, this supplies exactly the coefficient matching required in (4.2).

The factor $(1-t)^h$ is useful analytically and should survive the period matching. Simply truncating $\Omega(P_r)$ to the required band would lose this factor. Instead, replace each monomial by a polynomial with the same low coefficients and the required zero at $t = 1$:

$$F_i(t) = (1-t)^h \left[\frac{t^i}{(1-t)^h} \right]_{<L} \quad (0 \le i < L), \qquad E_r = \sum_{i=b}^{L-1} B_i(P_r) F_i. \tag{4.5}$$

Here $[\ ]_{<L}$ means Taylor truncation. Each F_i agrees with t^i below degree L and is divisible by $(1-t)^h$. Thus E_r has the right coefficients and retains the endpoint zero. We call this choice of E_r the *canonical completion*.

The actual entries are now explicit:

$$\begin{aligned} \mathcal{M}_{rj}(\lambda) &= M(P_r, Q_j) - \tfrac{3}{2} Z(D_r, Q_j) + 24\lambda Z(E_r, Q_j) \\ &= a_{rj}(\lambda) + b_{rj}\theta_\lambda, \end{aligned}$$

where $a_{rj}(\lambda)$ is the rational part and

$$b_{rj} = 4 \sum_{i=b}^{L-1} [s^i] Q_j(s)\, B_i(P_r).$$

Define $M_{N,\lambda}(X) = (a_{rj}(\lambda) + b_{rj}X)_{r,j<n}$. Its determinant is the canonical determinant polynomial. The endpoint zeros are already built into this matrix; the integer adjustments needed for the denominator estimate are introduced next.

4.3 High completions and low arithmetic corrections

There are two different freedoms in completing these coefficients. Changing the terms above degree $L - 1$ under the same degree and endpoint conditions has no effect on the determinant. Changing terms below degree b can change it, and will be needed for the arithmetic estimate. The following proposition explains the first assertion.

Proposition 4.1. *Let E_1, E_2 be linear functions of P, each divisible by $t^b(1-t)^h$, of degree at most $H-1$, and agreeing with $\Omega(P)$ in degrees $b, \dots, L-1$. Their mixed determinants are identical, as polynomials in the target and in λ.*

Proof. The difference belongs to $V = t^L(1-t)^h\mathbb{Q}[t]_{<g}$. Pair rows r and $2g-r$ for $0 \le r < g$. Their P polynomials agree and their D polynomials have opposite signs. Subtracting the rows leaves the rational row $3Z(D_{2g-r}, Q_j)$, independent of the target and of the completion. The polynomials D_{2g-r} form a basis of V: after removing $(1-t)^h$, their distinct lowest degrees are $L, \dots, C-1$. Every change of completion therefore adds a combination of these fixed rows to the remaining rows. All these row operations preserve the determinant. □

For the second freedom, set $\lambda = 1$ and $\theta = G+\pi^2$. Abbreviate $M_{N,1}(X)$ to $M_N(X) = A_N + XB_N$. Replace E_r by

$$E_r + \sum_{i=0}^{b-1} U_{ri}F_i, \qquad U \in \mathbb{Z}^{n\times b}.$$

The added moments are rational, because the low F_i have zero coefficients in the column band. With

$$W_{ij} = 24Z(F_i, Q_j), \qquad P_{N,U}(X) = \det(M_N(X) + UW), \tag{4.6}$$

we obtain a polynomial of degree at most n with rational coefficients. The entries of U will vary independently; Proposition 4.1 does not make these low corrections disappear. Thus two changes have different jobs. The canonical completion restores the endpoint zero while matching the target. The additional matrix UW changes only the rational parts of the entries, allowing us to impose prime divisibility conditions. Its real effect on the determinant is the unresolved part of the argument.

4.4 One arithmetic bound for an entire congruence class

It is useful to impose the denominator savings first and leave a whole congruence class of corrections available for the real estimate. The following result is the uniform conclusion of [R01].

Theorem 4.2. *There are integer matrices U_N^0 and positive integers a_N such that, with*

$$m_N = 2^C \prod_{2\sqrt{H}<p\le H,\ p\ odd} p^2,$$

every $U = U_N^0 + m_N Z$, $Z \in \mathbb{Z}^{n\times b}$, has entries divisible by 2^C and satisfies $a_N P_{N,U}(X) \in \mathbb{Z}[X]$. Moreover

$$\begin{aligned} \log a_N &\le (A_{\mathrm{arith}} + o(1))n^2, \\ A_{\mathrm{arith}} &= \frac{33079}{18432} + \frac{505}{4608}\log 2 = 1.87061400307785858968\dots, \\ \log m_N &\le \left(\frac{63\log 2 + 130}{48} + o(1)\right) n. \end{aligned} \tag{4.7}$$

The same a_N works for every integer Z, regardless of its size. One may choose $|U_{ri}^0| \le m_N/2$.

Here are the valuation arguments behind the constant. They also identify which parts are inherited from the moment construction. At 2, the smoothing identities of [5, (22)–(27)] split each raw column into a term with valuation $C - O(\log H)$, a multiple of its polynomial-part coefficient column, and a D term. Denote that multiple by $e_1(X)$: it is the source's fixed degree-one polynomial

over the 2-adic numbers, whose coefficient valuations are bounded below independently of N. Only this valuation bound is needed here.

To keep the column indices visible, let $\mathbf{\Omega} = (B_i(P_r))_{0\le r<n,\ b\le i<L}$ be the n by $(L-b)$ coefficient matrix of the polynomial parts. Multiplying it on the right combines these coefficient columns. For the mixed construction, its original contribution and the canonical completion combine as

$$\mathbf{\Omega}(e_1(X)I + 24Z_F), \qquad (Z_F)_{ij} = Z^0(F_i, j), \quad b \le i, j < L.$$

All entries of Z_F have valuation at least $-2\lfloor \log_2 H \rfloor$. The low correction, being divisible by 2^C, joins the smoothed term. This grouping retains the coefficient-band index when the determinant is expanded.

Write the coefficients of $P_r/(1-t)^h$ and $D_r/(1-t)^h$ in descending degrees as vectors p_u, d_u. The Chebyshev recurrence gives their entry valuations at least $u-1$. The scalar coefficient multiplying p_u in band position i of $\mathbf{\Omega}$ has valuation at least $u+i-H+1$. If a determinant term uses v columns from the $\mathbf{\Omega}$ group and w from the D group, alternation forces distinct band indices and distinct indices within each coefficient family. Summing their least possible values gives

$$v_2(P_{N,U}) \ge \min_{v,w} \left\{ -(H-b)v + \tfrac{3}{2}v^2 + \tfrac{1}{2}w^2 + C(n-v-w) \right\} - O(n \log H).$$

The minimum is over integers $v, w \ge 0$ with $v+w \le n$ and has $w = n-v$. With $(H-b-n)/n = 5/24$, it is $-505n^2/4608 - O(n \log n)$. The two alternations are responsible for the quadratic saving.

For odd $p > 2\sqrt{H}$, the moment reductions in [5, Section 3] have at most two denominator layers. Multiplying the low moment matrix by p^2 exposes a unit minor. If that minor has rank r, solving for the low coefficients modulo p^2 kills r columns and leaves cost at most $2(n-r)$. The monomial part of F_i supplies the minor; its endpoint tail has rank at most h. This gives ranks $b-h$ for $11N < p \le H/2$ and b for $H/2 < p \le 145N/4$.

For $145N/4 < p \le 52N$, use the b raw columns $j = p, \dots, p+b-1$. Their low block reduces to $24I_b$ after multiplication by p^2. Put $K = L-p$, $J = H-p$. After killing this block, $K+J-2b$ columns retain double poles, and the remaining simple-pole residues span at most b dimensions. The cost is therefore $2(K+J-2b)+b = 227N - 4p$. For $p > 52N$ no correction is required; the same residue count is

$$2\big(K_+ + (J-b)_+\big) + (\min(b, J) - \max(0, K))_+.$$

These calculations give the following piecewise cost $Nd(p/N)$. The entry $96N$ includes the range where no useful rank saving is used. For $2\sqrt{H} < p \le 11N$, choose the zero correction modulo p^2. This incurs the full cost $2n = 96N$ and will also supply the divisibility needed for nonvanishing.

$x = p/N$	$d(x)$	$x = p/N$	$d(x)$
$(0, 11]$	96	$(145/4, 52]$	$227 - 4x$
$(11, 65/2]$	86	$(52, 58]$	$182 - 3x$
$(65/2, 145/4]$	82	$(58, 59]$	$124 - 2x$
		$(59, 65]$	$65 - x$

The integral of d is $33079/8$. The prime number theorem, followed by division by $n^2 = 2304N^2$, gives $33079/18432$. Small odd primes cost $O(n\sqrt{H} \log H) = o(n^2)$; primes above H cost nothing. The Chinese remainder theorem combines the local choices with divisibility by 2^C. This proves Theorem 4.2, using the stated moment reductions. Their exact finite checks are indexed in Appendix B.

4.5 Nonvanishing under a rationality assumption

The arithmetic congruences also permit an independent nonvanishing argument. This matters because a sequence of zero integers gives no contradiction, however good its upper bound. At a prime scale $N = p > 260$, we have $H = 65p$ and $2\sqrt{65p} < p \le 11p$. Thus p lies in the range where the prescribed correction is zero modulo p^2. Since $p^2 \mid m_N$, every matrix $U = U_N^0 + m_N Z$ in the arithmetic class satisfies $p^2 \mid U$. This is the hypothesis used in the following proposition.

Proposition 4.3. *Fix rational λ. Outside a finite set of primes, at scale $N = p$ the mixed determinant $P_{N,\lambda,U}(X) = \det(M_{N,\lambda}(X) + UW)$ with $p^2 \mid U$ satisfies*

$$p^{96p} P_{p,\lambda,U}(X) \in \mathbb{Z}_{(p)}[X], \qquad p^{96p} P_{p,\lambda,U}(X) \bmod p \text{ is a nonzero constant.}$$

Consequently, for every fixed rational x, these determinants are nonzero at x for all sufficiently large primes. For $\lambda = 1$, every correction in Theorem 4.2 is allowed.

The proof in [R04] reduces the unbounded sizes to three polynomial seed determinants. To specify them, take the scale $N = 1$, allow the extra row $r = 48$, and form the 49 by 48 rational matrix

$$T_\eta(r,k) = \sum_{j=0}^{4} (-1)^j \binom{4}{j} \left\{ M^0(P_r, 7+k+j) - \tfrac{3}{2} Z^0(D_r, 7+k+j) + 24\eta Z^0(E_r, 7+k+j) \right\}.$$

Let $T_{\eta,0}$ use rows $0, \dots, 47$ and $T_{\eta,1}$ rows $1, \dots, 48$, and set $F_\sigma(Y) = \det(T_{Y,0} + \sigma T_{Y,1})$ for $\sigma = 0, 1, -1$. The paired rows remove four powers of Y, so each degree is at most 44. Exact finite-field calculations give

σ	Prime	Degree / leading coefficient	Number of roots
0	103	44 / 82	0
1	107	44 / 57	0
-1	103	44 / 5	0

This excludes every rational root. If its denominator is prime to the listed prime, reduction gives a finite-field root. If the denominator is divisible by that prime, homogenization to degree 44 leaves the nonzero leading term. The degree bound is needed for this second case.

To see why these are the correct seeds, put $\epsilon = (-1)^{(p-1)/2}$ and $K_p(t) = (1-t^2)^{(p-1)/2}$. The Frobenius identity

$$(1-t^{-2})^{-1/2} = \epsilon t^{1-p} K_p(t) (1-t^{-2p})^{-1/2} \pmod p$$

transfers the coefficient matching and its endpoint completion from scale p to scale one. The source moment digit reductions then give a block matrix

$$A_*^{\mathsf{T}} \otimes T_{\epsilon\lambda,0} + B_*^{\mathsf{T}} \otimes T_{\epsilon\lambda,1}, \qquad B_* = A_* \Pi,$$

where A_* is invertible and $\Pi e_0 = 0$, $\Pi e_i = e_{p-i}$ for $1 \le i < p$. Thus its determinant, up to a unit and a sign, is

$$F_0(\epsilon\lambda) F_1(\epsilon\lambda)^{(p-1)/2} F_{-1}(\epsilon\lambda)^{(p-1)/2}.$$

The target coefficient and the correction vanish in this reduction: the former is integral before scaling, and $p^2 \mid U$ while $v_p(W_{ij}) \ge -2$. Excluding the finitely many numerator and denominator primes of the six specialized seed values proves the proposition.

Under the hypothesis $\theta = a/d \in \mathbb{Q}$, we now have nonzero integers

$$d^n a_N P_{N,U}(\theta)$$

along large prime scales, for every admissible correction. The remaining task is analytic: choose those corrections so that their determinants decay faster than $\exp(-A_{\mathrm{arith}} n^2)$.

5 A certified decay estimate for the canonical determinant

5.1 What the estimate supplies

Let $\Delta_N = \det M_N(\theta)$, where $\theta = G + \pi^2$ and the canonical completion is (4.5). The analytic investigations [R07], [R14], [R18]–[R20] and [R23] culminate in the following bound.

Theorem 5.1. *For $n = 48N$,*

$$\limsup_{N\to\infty} \frac{\log|\Delta_N|}{n^2} \le -1.87063289244902359295.$$

This decay constant exceeds A_{arith} in Theorem 4.2 by more than 0.00001888937116500327.

The canonical completion retains the endpoint zero. It does not impose the integer congruences of Theorem 4.2. The theorem therefore closes one analytic gap: it gives enough decay before those additional corrections are made. Section 5.6 states the estimate still needed to complete the contradiction.

The small margin makes the treatment of volume important. Separate row norms lose too much information about near dependence. We retain Gram determinants, first for the main rows and then for their endpoint repair. A Gram determinant measures the squared volume spanned by the vectors, and thus records this dependence.

The proof has three quantitative parts. Gram Cauchy–Schwarz bounds $|\Delta_N|$ by the square root of the product of the row and column Gram determinants. The row estimate splits into the main rows and the repair required by the endpoint zero. Thus the bound for $n^{-2}\log|\Delta_N|$ is half the sum of the first three contributions below; changing its sign gives the decay constant.

Main row logarithmic rate	$-0.19795874093499631465\ldots$
Endpoint repair, rounded upper bound	0.00498
Column logarithmic rate	$-3.54828704396305087125\ldots$
Canonical decay, lower bound	$1.87063289244902359295\ldots$
Arithmetic cost, upper bound	$1.87061400307785858968\ldots$

The resulting margin is about 1.89×10^{-5} on the n^2 scale. The next four subsections derive the volume decomposition and its three bounds. The final subsection measures how much of this margin the later integer corrections may consume.

5.2 Separate the row and column volumes

Put

$$d\nu(t) = \frac{\mathrm{d}t}{\pi\sqrt{1-t^2}} \quad (-1 < t < 1), \qquad d\sigma(t) = \frac{\sqrt{1-t^2}}{\pi}\,\mathrm{d}t \quad (0 < t < 1),$$

and write $w(t) = t^b(1-t)^h$, $R_r = P_r/w$, $S_r = D_r/w$, $V_r^{\mathrm{comp}} = E_r/w$. For a column polynomial Q, set $\mathcal{K}Q(t) = \int_0^1 Q(s)/(1-ts)\,\mathrm{d}s$. The actual mixed entry is the inner product $\pi\langle \mathcal{A}_r, Y_j\rangle_\nu$, where

$$\begin{aligned}\mathcal{A}_r(t) &= |t|R_r(t) + \mathbf{1}_{t>0}\sqrt{1-t^2}\,(-\tfrac{3}{2}S_r(t) + 24V_r^{\mathrm{comp}}(t)),\\ Y_j(t) &= w(t)\mathcal{K}Q_j(t).\end{aligned}$$

Gram Cauchy–Schwarz consequently gives

$$|\Delta_N| \le \pi^n \sqrt{\det G_{\mathcal{A}} \det G_Y}. \tag{5.1}$$

The polynomial-part operation has a useful norm bound. Chebyshev orthogonality on the full semicircle gives $\|\Omega(R)\|_\sigma \le \|R\|_\nu$; restricting to the positive half can only decrease the norm. Multiplication by $(1-t)^h$ does not commute with Ω, however. If $a_s(R) = \int x^s R(x)\,\mathrm{d}\nu(x)$, the exact defect is

$$\Omega((1-t)^h R) = (1-t)^h\Omega(R) + J_h(R), \qquad J_h(R) = \sum_{\ell=1}^{h}(-1)^\ell \binom{h}{\ell}\sum_{s=1}^{\ell} a_s(R)t^{\ell-s}. \tag{5.2}$$

For $h=2$, the defect is $(a_2 - 2a_1) + a_1 t$. Thus its dimension is only h, even though the original row degrees grow much faster.

Let $\ell = 56N-1$, $M = 52N$, and define

$$\begin{aligned}R_r(t) &= t^\ell T_{|r-g|}(1/t),\\ V_r(t) &= \operatorname{sgn}(r-g)t^{\ell-1}U_{|r-g|-1}(1/t), \qquad V_g = 0,\\ C_k(t) &= \sum_{j=k}^{M-1}\binom{h+j-k-1}{j-k}t^j, \qquad 0 \le k < h,\\ a_k(x) &= \sum_{a=k+1}^{h}(-1)^a\binom{h}{a}x^{a-k}.\end{aligned}$$

Let H_N be the Gram matrix of (R_r, V_r) in $L^2(\nu)\oplus L^2(\sigma)$, let $J_{rk} = \langle R_r, a_k\rangle_\nu$, and let B^{rep} be the Gram matrix of the C_k in $L^2(\sigma)$. The kernel of the repair is $\sum_k a_k(x)C_k(t)$. Formula (5.2) makes its role explicit: since $J_{rk} = [t^k]J_h(R_r)$ and $C_k = [t^k/(1-t)^h]_{<M}$, take

$$\widehat{E}_r = w\left(\Omega(R_r) + \sum_{k=0}^{h-1} J_{rk}C_k\right).$$

This completion agrees with $\Omega(P_r)$ in degrees $b,\dots,L-1$, is divisible by w, and has degree at most $H-1$. Proposition 4.1 therefore allows us to replace E_r by $\widehat{E}_r$ without changing Δ_N. Since $S_r = tV_r$, the resulting row function is

$$\begin{aligned}\widehat{\mathcal{A}}_r(t) = |t|R_r(t) &+ 24\mathbf{1}_{t>0}\sqrt{1-t^2}\,\Omega(R_r)(t)\\ &- \tfrac{3}{2}\mathbf{1}_{t>0}\sqrt{1-t^2}\,tV_r(t)\\ &+ 24\mathbf{1}_{t>0}\sqrt{1-t^2}\sum_{k=0}^{h-1}J_{rk}C_k(t).\end{aligned}$$

Use these rows in (5.1), with $G_{\widehat{\mathcal{A}}}$ in place of $G_{\mathcal{A}}$. The first line defines the bulk map from R_r; its norm is at most $1+24=25$ by the contraction bound above. The maps in the next two lines have norms

at most 3/2 and 24, respectively. The input triples $(R_r, V_r, \sum_k J_{rk} C_k)$, in $L^2(\nu) \oplus L^2(\sigma) \oplus L^2(\sigma)$, have Gram matrix

$$K_N = H_N + J B^{\mathrm{rep}} J^{\mathsf{T}},$$

so the squared norm of the combined map is at most $25^2 + (3/2)^2 + 24^2 = 4813/4$. Consequently,

$$\det G_{\widehat{\mathcal{A}}} \le (4813/4)^n \det K_N, \qquad J^{\mathsf{T}} H_N^{-1} J \le (\langle a_i, a_j \rangle_\nu)_{i,j<h}. \tag{5.3}$$

The second inequality is the contraction of an orthogonal projection. The paired rows and their distinct lowest powers show that H_N is positive definite. All constant factors raised to n disappear after division of logarithms by n^2.

5.3 The main rows: retain the spectrum of the row operator

The paired rows split the main volume into an explicit Jacobi volume and the extra volume contributed by a single operator. Put $a = 12N$, $m = 44N$, and

$$\mathcal{E} = t^{\ell-g}\mathbb{R}[t]_{<g}, \qquad \mathcal{F} = t^a \mathbb{R}[t]_{<m}.$$

On the basis $R_d = t^\ell T_d(1/t)$, $0 \le d < m$, define $\mathcal{L}R_d = t^{\ell-1} U_{d-1}(1/t)$, with $\mathcal{L}R_0 = 0$. Let $P_{\mathcal{E}}$ be orthogonal projection in $L^2(\sigma)$ and $\mathcal{T} = (I - P_{\mathcal{E}})\mathcal{L}$. Subtracting the paired rows gives $(0, -2V_d)$, $1 \le d \le g$. Taking their Schur complement therefore yields

$$\det H_N = 4^g \det Z_N \det A_N^{\mathrm{row}} \det(I + \mathcal{T}^*\mathcal{T}), \tag{5.4}$$

where Z_N and A_N^{row} are the Gram matrices of these V_d and of $R_0, \dots, R_{m-1}$, respectively.

The first two factors can be evaluated through monic Jacobi norms. For weight $x^u(1-x)^v$ on $[0,1]$, that norm is

$$\frac{j!\,\Gamma(j+u+1)\Gamma(j+v+1)\Gamma(j+u+v+1)}{(2j+u+v+1)\Gamma(2j+u+v+1)^2}.$$

The triangular leading coefficients of the Chebyshev bases contribute $2^{(m-1)(m-2)}$ and $2^{g(g-1)}$. Replacing the remaining bounded weight factors changes the logarithm by $O(n)$. Stirling's formula then gives the limiting base contribution

$$\begin{aligned} b_0 &= \tfrac{61}{72}\log 2 + 2\mathcal{I}(1/4, 0, 11/24) + \mathcal{I}(13/6, 0, 1/12) \\ &= -0.243039538134996314659\ldots, \\ \mathcal{I}(u, v, T) &= \int_0^T \{f(t) + f(t+u) + f(t+v) + f(t+u+v) - 2f(2t+u+v)\}\,\mathrm{d}t, \end{aligned}$$

Here $f(x) = x \log x$, with $f(0) = 0$.

We still need the last factor in (5.4). If $s_1 \ge \cdots \ge s_m \ge 0$ are the singular values of $\mathcal{T}$, then

$$\log\det(I + \mathcal{T}^*\mathcal{T}) = \sum_{j=1}^{m} \log(1 + s_j^2).$$

Bounding every s_j by s_1 would charge the worst direction m times. Instead, we approximate $\mathcal{T}$ by operators of successively larger rank. The error after a rank-l approximation bounds s_{l+1}, allowing the later, smaller directions to contribute less to this sum.

For $R \in \mathcal{F}$, put $q_R(x) = x^\ell R(1/x)$. The useful exact representation is

$$\mathcal{L}R(t) = \left[\frac{R(t)}{\sqrt{1-t^2}}\right]_{<\ell} = \frac{R(t)}{\sqrt{1-t^2}} - t^\ell \int \frac{q_R(x)}{1-xt}\,\mathrm{d}\nu(x).$$

The first term is a contraction from $L^2(\nu)$ to $L^2(\sigma)$. Approximate the second term by rational functions. Choose $g+l$ zeros $z_i \in (0,1)$ and l poles $y_j \in (-1,1)$, and set

$$P(t)=\prod_i (t-z_i), \quad Q(t)=\prod_j (1-y_j t), \quad W_*(t)=t^{\ell-g}P(t)/Q(t), \quad B_*(x)=\frac{\prod_j (x-y_j)}{\prod_i (1-z_i x)}.$$

Polynomial division gives

$$\frac{t^\ell}{1-xt}=\frac{t^{\ell-g}v_x(t)}{Q(t)}+\frac{W_*(t)B_*(x)}{1-xt}, \qquad \deg v_x < g+l.$$

The first space contains $\mathcal{E}$, so it has rank at most l after projection. This converts a uniform bound on the second term into a bound on the $(l+1)$st singular value.

The necessary evaluation estimate is

$$|q_R(x)| \le C_N e^{n\mathcal{E}_0(x)}\|R\|_\nu, \qquad C_N=O(n),$$

where, with $\tau=11/24$ and $\alpha=1/4$,

$$\mathcal{E}_0(x)=\inf_{u>0}\{\tau\log(1+u)+(\tau+\alpha)\log(1+(1-x^2)u)-\tau\log u\}.$$

At $x=\pm1$ its value is zero. To obtain the estimate, expand in the orthonormal Jacobi basis and use the positive coefficient sum $\sum_i \binom{K}{i}\binom{K+a}{i}(1-x^2)^i$, $K=22N$. It is the coefficient of u^K in $(1+u)^K(1+(1-x^2)u)^{K+a}$; bounding that coefficient at positive u and minimizing gives the displayed field. The evaluation norms supply, for example, $C_N=\sqrt{\pi m(2K+a+2)}$.

For $l/n \to r$, describe the zeros and poles by positive measures p, q of masses $1/12+r, r$. The two potentials to bound are

$$\mathcal{U}(t)=\tfrac{13}{12}\log t+\int \log|t-z|\,\mathrm{d}p(z)-\int \log(1-yt)\,\mathrm{d}q(y),$$
$$\mathcal{V}(x)=\mathcal{E}_0(x)+\int \log|x-y|\,\mathrm{d}q(y)-\int \log(1-zx)\,\mathrm{d}p(z).$$

If their upper bounds sum to $e(r)\ge 0$, the singular-value estimate is $s_{l+1}(\mathcal{T}) \le 1+\exp(n(e(r)+o(1)))$. The accompanying rational atomic measures certify the following values.

r	0	.02	.04	.06	.08	.10
$e(r)$	.43055090	.32535779	.24085716	.16822647	.10576227	.05324220
r	.12	.14	.16	.18	.20	.22
$e(r)$	.01233031	.00066131	.00029828	.00023083	.00019427	.00015768
r	.24					
$e(r)$	.00012891					

Convex interpolation of the measures proves the linear interpolant is an upper bound between nodes. After .24, adding equal masses at $1/2$ to both measures cannot increase either potential, since $|(x-1/2)/(1-x/2)| \le 1$ on $[-1,1]$. This extends the final bound to $r=11/12$. Rational masses can be rounded to integer multiplicities with only $O(1)$ residual nodes. Upper bounds for the logarithmic potentials persist in this passage; zeros make the potential smaller, not larger. At an index j with $j/n \approx r$, the contribution $\log(1+s_j^2)$ is bounded on the leading scale by $2ne(r)$.

Summing these contributions over the $m = 11n/12$ singular values gives the integral $2\int_0^{11/12} e(r)\,\mathrm{d}r$ after division by n^2. Together with the Jacobi contribution in (5.4), this gives

$$\limsup \frac{\log\det H_N}{n^2} \le b_0 + 2\int_0^{11/12} e(r)\,\mathrm{d}r = b_0 + \frac{112701993}{2500000000} = -0.19795874093499631465\ldots. \tag{5.5}$$

The certificate uses 192-bit interval arithmetic and a global curvature bound on each subinterval; it checks both potentials at all 13 nodes, rather than sampling their maxima.

5.4 The repair: keep the negative interaction between degrees

The repair in (5.2) is already part of the canonical matrix. It is distinct from the later arithmetic correction UW. The kernel simplifies to

$$K(x,t) = -x\sum_{j=0}^{h-1}(1-x)^j \sum_{r=0}^{M-1}\binom{r+j}{j}t^r.$$

Let $A_{ij} = \langle -x(1-x)^i, -x(1-x)^j\rangle_\nu$ and $B_{ij} = \sum_{r=0}^{M-1}\binom{r+i}{i}\binom{r+j}{j}$. The elementary bound $\|\sum c_r t^r\|_\sigma^2 \le M\sum c_r^2$ and (5.3) give

$$\det K_N/\det H_N \le \det(I_h + MAB).$$

For a k-element set I, let A_I denote its principal minor. Cauchy–Binet and Gram Cauchy–Schwarz imply

$$e_k(AB) \le \binom{h}{k}^2 \max_{|I|=k}\det A_I \det B_I. \tag{5.6}$$

Here $e_k(AB)$ is the kth elementary symmetric function of the eigenvalues of AB, so $\det(I_h+MAB) = \sum_{k=0}^h M^k e_k(AB)$. This keeps the interaction between the selected degrees. Replacing the minors by separate vector norms would discard that saving.

Put $L_* = M + h = 27h$ and $\mathcal{H}_I = (1/(i+j+1))_{i,j\in I}$. The two comparisons needed are

$$\det A_I \le h^{2k}2^{2\sum_I j}\det\mathcal{H}_I, \qquad \det B_I \le (Mh^2)^k\frac{L_*^{2\sum_I j}}{\prod_I j!^2}\det\mathcal{H}_I. \tag{5.7}$$

For the first, change variables $y = 1-x$ and use $x^2 \le 1$. The polynomial evaluation inequality $\sup_{[0,L_0]}|f|^2 \le (h^2/L_0)\int_0^{L_0}|f|^2$ for $\deg f < h$, obtained from shifted Legendre polynomials, bounds the arcsine Gram by the uniform Gram on $[0,2]$.

For the second comparison, the factorial Schur tableau identity [1, Theorem 2 and (7)–(8), (14)] gives, for increasing nonnegative r_a and increasing $j_b < h$,

$$0 \le \det\left[\prod_{v=1}^{j_b}(r_a+v)\right] \le \det[(r_a+h)^{j_b}].$$

Indeed, divide by the Vandermonde and use the partition $\lambda_i = j_{k+1-i} - k + i$. Every tableau factor has shift between 1 and $j_k < h$, so replacing it by h increases every nonnegative summand. Multiplication by the Vandermonde proves the comparison. Apply it to each evaluation minor of B_I, divide column j by $j!$, and then use the polynomial evaluation inequality on $[0, L_*]$. This proves (5.7). The binomial comparison is a comparison of determinants; no entrywise matrix ordering is asserted.

The saving from these minors is visible with just two degrees. For $I = \{0, 1\}$,

$$\mathcal{H}_I = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{pmatrix}, \qquad \det \mathcal{H}_I = \frac{1}{12}, \qquad (\mathcal{H}_I)_{00}(\mathcal{H}_I)_{11} = \frac{1}{3}.$$

The determinant is four times smaller than the product of the separate squared norms. For general $i < j$, the corresponding ratio is

$$\frac{\det \mathcal{H}_{\{i,j\}}}{(\mathcal{H}_{\{i,j\}})_{ii}(\mathcal{H}_{\{i,j\}})_{jj}} = \left(\frac{j-i}{i+j+1}\right)^2.$$

Nearby powers are nearly dependent, and their joint volume records this. Both comparisons in (5.7) contain the same Cauchy minor, so their product contributes the factor $((j-i)/(i+j+1))^4$. These factors reduce the bound; their logarithms give the negative interactions retained below.

The Cauchy determinant formula now gives

$$\det K_N / \det H_N \le (4M^2h^4)^h \max_{I \subset \{0,\dots,h-1\}} \mathcal{W}_I,$$
$$\mathcal{W}_I = \prod_{j \in I} \left(\frac{(2L_*)^j}{j!}\right)^2 \prod_{i<j \in I} \left|\frac{i-j}{i+j+1}\right|^4.$$

Use the midpoint lattice $x_j = (j + 1/2)/h$. Stirling's formula turns the first product into the field $w_0(x) = 2x(\log(54/x) + 1)$. The pair interaction is $G_0(x, y) = \log|(x - y)/(x + y)|$.

Its Chebyshev expansion is $G_0 = -4\sum_{k\ge1,\ k\ \text{odd}} T_k(x)T_k(y)/k$. To make the passage to a uniform bound legitimate, damp by ρ^k, $0 < \rho < 1$. The damped kernel is

$$G_\rho(x, y) = \tfrac{1}{2}\log\frac{D_0 - D_1}{D_0 + D_1}, \quad D_0 = (1-\rho^2)^2 + 4\rho^2(x^2+y^2), \quad D_1 = 4\rho(1+\rho^2)xy.$$

On $[0, 1]^2$, $G_0 \le G_\rho$. After cross multiplication the required difference is $4xy(1-\rho)^2((1+\rho)^2 - 2\rho(x^2+y^2)) \ge 0$. To expose the squares in this interaction, define the averaged moments

$$m_k(I) = \frac{1}{h}\sum_{j \in I} T_k(x_j), \qquad k \ge 1 \text{ odd}.$$

The field contribution is at most $h\sum_{j\in I} w_0(x_j)$, using $j! \ge (j/e)^j$ and the monotonicity of w_0 on $[0, 1]$. Replacing G_0 by G_ρ and inserting the diagonal in the pair sum gives

$$\frac{\log \mathcal{W}_I}{h^2} \le \frac{1}{h}\sum_{j\in I} w_0(x_j) - 8\sum_{k\ \text{odd}} \frac{\rho^k m_k(I)^2}{k} + \frac{4}{h}\log\frac{1+\rho}{1-\rho}.$$

Here the last term removes the inserted diagonal: $G_\rho(x, x) \ge -2\log((1 + \rho)/(1 - \rho))$. It tends to zero for fixed ρ as $h \to \infty$.

Choose any finitely supported real sequence v_k on the positive odd indices. The elementary inequality

$$-m_k(I)^2 \le v_k^2 - 2v_k m_k(I)$$

replaces each negative square by a term linear in the moments. After substituting their definition, site x_j contributes the field $w_0(x_j) - 16\sum_{k\ \text{odd}} \rho^k v_k T_k(x_j)/k$ if it belongs to I, and zero otherwise. Its contribution is therefore at most the positive part of this field. Summing over all sites removes

the dependence on I. First let $h \to \infty$, so the full lattice sum becomes an integral, and then let $\rho \uparrow 1$. Since $n = 24h$, the resulting bound is

$$\limsup \frac{\log(\det K_N / \det H_N)}{n^2} \le \frac{1}{576} \left\{ 8 \sum_{k \text{ odd}} \frac{v_k^2}{k} + \int_0^1 \left(w_0(x) - 16 \sum_{k \text{ odd}} \frac{v_k T_k(x)}{k} \right)_+ \mathrm{d}x \right\}. \quad (5.8)$$

This order of limits avoids using an undamped logarithmic series on the diagonal. It also retains the occupancy restriction on I.

The supplied trial uses odd indices through 79 and rational coefficients with denominator 10^{12}. A 512-bit calculation on 16384 intervals bounds the positive-part integral by 0.85979724477769159413. On each interval the polynomial's second derivative is bounded by $\frac{16}{3} \sum |v_k| k(k^2 - 1)$; the field's second derivative is negative. An affine upper bound, with this quadratic error added, can be integrated exactly even where it crosses zero. The result is

$$\limsup \frac{\log(\det K_N / \det H_N)}{n^2} \le 0.004978210597652675202 < 0.00498. \quad (5.9)$$

5.5 The columns: an energy bound with certified maxima

Expanding the column Gram determinant twice with the Cauchy determinant formula expresses it as a positive integral:

$$\det G_Y = \frac{1}{(n!)^3} \int \frac{V(t)^2 V(s)^2 V(u)^2 \prod_i w(t_i)^2 s_i^b (1 - s_i)^q u_i^b (1 - u_i)^q}{\prod_{i,j} (1 - t_i s_j)(1 - t_i u_j)} \prod_i \mathrm{d}\nu(t_i)\, \mathrm{d}s_i\, \mathrm{d}u_i,$$

where V denotes the Vandermonde, $t_i \in (-1, 1)$ and $s_i, u_i \in (0, 1)$. This positivity permits a uniform bound on the integrand. Make the substitution $t = 2x/(1 + x^2)$, and put

$$\beta = 7/48, \quad \eta = 1/24, \quad \gamma = 1/12,$$

$$W_0(x) = \beta \log|x| + 2\eta \log(1 - x) - (\beta + \eta) \log(1 + x^2), \qquad Z_0(s) = \beta \log s + \gamma \log(1 - s).$$

For any finite sequences p_k, v_k, the energy calculation gives

$$\begin{aligned} \limsup \frac{\log \det G_Y}{2n^2} \le{}& (\beta - 1) \log 2 + \sum_k \frac{p_k^2}{k} + \frac{1}{2} \sum_k \frac{v_k^2}{k} \\ &+ \sup_{-1 \le x \le 1} \left(W_0(x) - 2 \sum_k \frac{p_k T_k(x)}{k} - \sum_k \frac{v_k x^k}{k} \right) \\ &+ \sup_{0 \le s \le 1} \left(Z_0(s) + 2 \sum_k \frac{v_k T_k(s)}{k} \right). \end{aligned} \quad (5.10)$$

For completeness, the square completion behind this estimate keeps three empirical measures, one each for x, s, u. Let A_k be the Chebyshev moments of the first, B_k its power moments, and C_k, D_k the Chebyshev moments of the other two. The interaction, apart from its constant, is

$$-\sum_k \frac{A_k^2}{k} - \frac{1}{4} \sum_k \frac{(B_k - 2C_k)^2 + (B_k - 2D_k)^2}{k}.$$

Applying $-a^2 \le p^2 - 2pa$ gives (5.10). To justify the expansion uniformly at endpoints and repeated points, use dampings $\tau = 1 - \varepsilon$ for the x Chebyshev terms, $\rho(x) = 1 - \varepsilon(1 - x)$ for its power terms,

and $\sigma(s) = 1 - \varepsilon\sqrt{1-s}$ for the s, u terms. The kernel comparison costs $O(\varepsilon)$ on the n^2 scale; diagonal errors are $O_\varepsilon(n)$, and the endpoint logarithms are absorbed by the positive exponents β, η, γ. Let $n \to \infty$ first, then $\varepsilon \downarrow 0$. This yields the displayed bound for the full integral, not just its interior.

For the rational degree-200 trial, the two maxima are bounded by

$$-\frac{1798431303}{5000000000}, \qquad -\frac{7455295417}{5000000000},$$

respectively. Clearing the denominators in the derivatives gives exact polynomials. Sturm counts give 96 roots on $(-1, 0)$, 89 on $(0, 1)$ for the first function, and 103 on $(0, 1)$ for the second. The certificate encloses all 288 roots, evaluates every critical value and the regular endpoint $x = -1$, and handles the other endpoints by their logarithmic limits. It proves

$$\limsup \frac{\log\det G_Y}{n^2} \le -3.54828704396305087125. \tag{5.11}$$

5.6 The margin and the estimate still missing

Combining (5.1), (5.3), (5.5), (5.9) and (5.11) proves Theorem 5.1. Taking half the sum of the three contributions listed at the start of the section gives the stated decay bound. The unrounded repair bound gives a decay constant of $1.87063378715019725535\ldots$. Its margin over the arithmetic cost is $0.00001978407233866567\ldots$.

Where $M_N(\theta)$ is invertible, the determinant lemma writes the effect of a later integer correction as

$$P_{N,U}(\theta) = \Delta_N \det(I_b + W M_N(\theta)^{-1} U).$$

For example, with the induced maximum-row-sum norm, it would suffice to choose admissible corrections along the prime scales such that

$$\limsup \frac{1}{n} \log(1 + \|W M_N(\theta)^{-1} U\|_\infty) < 0.00012952711656002246. \tag{5.12}$$

Indeed, the logarithm of the extra determinant is at most b times this logarithm, and $b/n = 7/48$. The unrounded repair allows the slightly larger threshold 0.00013566221032227892. These are sufficient conditions, not estimates that have been proved. One could also bound the corrected determinant directly and avoid the inverse.

Appendix B records the independent audit's scope. The next section examines ways to obtain the missing correction estimate and the limitations established for those approaches.

6 What integer corrections can attain

The remaining question is how much the arithmetic congruences can increase the determinant. We examine two ways to control that increase: describe the least attainable nonzero value under the rationality assumption, or construct corrections with a sufficiently small real effect. The first approach leads to a row-lattice covolume; the second leads to rounding in the correction's output space. We then test polynomial kernel inverses as a way to bound the covolume, and explain why successful finite searches do not settle the uniform question.

6.1 An arithmetic progression of determinant values

The correction matrix has many entries, but its determinant values have a simple exact description at a rational target. This lets us replace a search over large integer matrices by a question about the step of one arithmetic progression. The reduction does not control the height of the matrices needed to attain those values.

We begin with a square congruence problem. If $T \in \mathbb{Z}^{b\times b}$ has Smith invariants $d_1 \mid \cdots \mid d_b$, including possible trailing zeros, and ℓ is a positive integer, then

$$\{\det(T+\ell K) : K \in \mathbb{Z}^{b\times b}\} = \det T + \ell \prod_{i=1}^{b-1} \gcd(\ell, d_i)\mathbb{Z}. \tag{6.1}$$

The empty product for $b = 1$ is one. To prove necessity, expand the determinant by the number of changed entries. At a prime with $v_p(\ell) = e$, the smallest possible valuation of a nonconstant term is $e + \sum_{i<b} \min(e, v_p(d_i))$. For attainment, pass to Smith form and choose the first $b-1$ rows upper bidiagonal, with diagonal d_i and superdiagonal ℓ. Their last-row cofactors are

$$c_j = (-1)^{b+j}\ell^{b-j} d_1 \cdots d_{j-1}, \qquad 1 \le j \le b.$$

Their gcd is $\prod_{i<b} \gcd(\ell, d_i)$. Varying the last row by ℓ times an integer row and using Bézout's identity attains every value in (6.1). For example, matrices congruent to $\operatorname{diag}(2, 6)$ modulo 4 have determinants exactly $12 + 8\mathbb{Z}$. The matrices $\left(\begin{smallmatrix} 2 & 4 \\ 4u & 6+4v \end{smallmatrix}\right)$ already attain them all.

For an invertible rational M and a rational b by n matrix W of rank b, put $R = WM^{-1}$. Suppose corrections are $U = U_0 + mZ$. A basis matrix H_Λ for the rational lattice $mR\mathbb{Z}^n$ gives

$$\det(M + UW) = \det M \det H_\Lambda \det\big(H_\Lambda^{-1}(I_b + RU_0) + K\big), \qquad K \in \mathbb{Z}^{b\times b}.$$

Every K is attainable because the columns vary independently. Clearing a denominator of the last fixed matrix and applying (6.1) proves that the determinant values form an arithmetic progression. Equivalently, its positive step is the rational gcd of the nonconstant coefficients of $\det(M + U_0W + mZW)$. These coefficients are mixed row minors; finite differences recover them from integer-input values. This is the exact-image result of [R16].

6.2 The combined row lattice determines the leading rate

Here is an intrinsic form that also allows a singular initial matrix. For finitely many rational numbers, their *rational content* means the positive generator of their integer span. It is computed by clearing a denominator and taking an ordinary gcd.

Theorem 6.1. *Let $M_0 \in \mathbb{Q}^{n\times n}$ and $V \in \mathbb{Q}^{b\times n}$, with $\operatorname{rank} V = b$ and $\operatorname{rank}\left[\begin{smallmatrix} M_0 \\ V \end{smallmatrix}\right] = n$. Choose a row basis H_Λ of the lattice generated by the stacked rows, with positive determinant, and put*

$$\delta = \det H_\Lambda, \qquad A = M_0 H_\Lambda^{-1}, \quad B = V H_\Lambda^{-1}, \quad a = \det A, \quad c = \gcd_{i,j} B_{ij}.$$

Then A, B are integer matrices and

$$\{\det(M_0 + ZV) : Z \in \mathbb{Z}^{n\times b}\} = \delta(a + c\mathbb{Z}), \qquad \gcd(a, c) = 1. \tag{6.2}$$

Moreover, δ is the rational content of all maximal minors of the stack. Both δ and c are independent of the chosen basis.

Proof. The rows of $[A;B]$ generate $\mathbb{Z}^n$, so this matrix has an integer left inverse. Cauchy–Binet shows that its maximal minors have gcd one. Multiplying back by H_Λ proves the assertion about δ.

Some $M_0 + Z_*V$ is invertible: a nonzero mixed minor is a nonzero coefficient of its determinant polynomial, and a nonzero polynomial cannot vanish at every integer tuple. Shift the initial matrix by this Z_* and apply the progression result above. Its step, divided by δ, is the gcd of maximal minors of $[A;B]$ using at least one row from B.

Fix a prime p, and put $e = \min v_p(B_{ij})$. If $e = 0$, extend a nonzero row of B modulo p to a basis using rows of $[A;B]$. This gives a unit mixed minor. If $e > 0$, then A is invertible modulo p. All mixed minors have valuation at least e, and the one-row replacements are entries of $B\,\mathrm{adj}(A)$, up to sign. Since $\mathrm{adj}(A)$ is a unit matrix over $\mathbb{Z}_p$, one has valuation exactly e. Thus the step is c, prime by prime. The same argument shows that no prime dividing c divides a. □

For example, take $M_0 = \mathrm{diag}(1/2, 1)$ and $V = (5/3, 0)$. The row lattice has basis $\mathrm{diag}(1/6, 1)$, so $\delta = 1/6$, $a = 3$, $c = 10$. The determinants are $(3 + 10z)/6$, and their least absolute value is $1/2$. The lattice covolume is not itself always attained.

If zero is unattainable in (6.2), its least absolute value μ satisfies

$$\delta \le \mu \le \delta c/2. \tag{6.3}$$

If $c = 1$, zero is attainable and the least *nonzero* absolute value is δ. We next show that c is subexponential on the n^2 scale for the moment matrices. This leaves δ as the quantity whose leading rate matters.

If an integer D clears every entry of M_0, V, then DH_Λ is integral. Writing $B = cB'$ gives $DV = cB'(DH_\Lambda)$, so

$$c \le D \max_{i,j} |V_{ij}|. \tag{6.4}$$

For the construction of Section 4, fix $x = u/d \in \mathbb{Q}$ in lowest terms and take $M_0 = M_N(x) + U_N^0 W$, $V = m_N W$. The bound on the low-moment rank for $65N/2 < p \le 145N/4$ gives $\operatorname{rank} W = b$ for all sufficiently large N, since that interval then contains a prime. Proposition 4.3 makes $M_N(x)$ invertible along large prime scales. Thus the rank hypotheses hold on the sequence used in the proposed contradiction. A common denominator

$$D = d\,2^{H+1}((2H+2)!)^2$$

clears all entries. Indeed, $Z^0(i,i) = -\sum_{j=1}^{i} j^{-2}$ and $Z^0(i,j) = (H_i - H_j)/(i-j)$ off the diagonal. The rational M moments satisfy the following recurrences, with $c_j = 4^{-j}\binom{2j}{j}$:

$$\begin{gathered} t_{2j} = 2/((2j+1)c_j), \quad t_{2j+1} = 0, \quad v_0 = w_0 = w_1 = 0, \quad v_1 = 2, \\ v_k = ((k-1)v_{k-2} + t_{k-2} + t_{k-1})/k, \qquad w_k = ((k-2)w_{k-2} + 2/(k-1))/(k-1), \\ M^0(i,0) = v_i, \quad M^0(0,j) = w_j, \qquad M^0(i,j) = M^0(i-1,j-1) - t_{i-1}/j. \end{gathered}$$

Expanding a recurrence introduces products of distinct integers no larger than $2H+2$; the second factorial covers the forcing terms. The polynomial-part projection has denominator dividing 2^H. All other row and column coefficients are integers, proving the claim.

The coefficient sum of F_i is at most 2^{L+2h}, the column filter has coefficient sum 2^q, and $|Z^0(i,j)| < 2$. Hence $\max |W_{ij}| \le 48\,2^{L+2h+q}$. Also $m_N \le 2^C (H!)^2$. Formula (6.4) gives

$$c \le 48d\,2^{H+1+C+L+2h+q}((2H+2)!)^2(H!)^2, \qquad \log c = O_x(n \log n).$$

Together with (6.3), this proves

$$\log \mu = \log \delta + O_x(n \log n) \tag{6.5}$$

whenever zero is unattainable. This is the all-size conclusion of [R03]. Under a hypothetical rational value $x = \theta$, a sufficiently strong estimate for this row-lattice covolume would finish the argument. No such estimate has been established.

6.3 A prime-by-prime comparison of the index loss

The covolume can be compared directly with the canonical determinant. At a rational x where $M = M_N(x)$ is invertible, put

$$\Lambda_1 = \mathrm{row}_{\mathbb{Z}}[M; W], \quad \Lambda_0 = \mathrm{row}_{\mathbb{Z}}[M + U_0 W; mW], \quad I = [\Lambda_1 : \mathrm{row}_{\mathbb{Z}}\, M], \quad J = [\Lambda_1 : \Lambda_0].$$

Then $\delta/|\det M| = J/I$. At $p \nmid m$, the local spans of Λ_0, Λ_1 agree, so the local contribution to J/I is nonpositive. If $e = v_p(m) > 0$ and u_i are the Smith invariants of U_0 modulo p^e, a local diagonalization gives

$$v_p(J/I) \le \sum_{i=1}^{b} (e - v_p(u_i))_+.$$

Indeed, multiplying the affected canonical rows by the indicated powers of p puts them in Λ_0; this bounds the loss of index. For the choices in Theorem 4.2, only primes $11N < p \le 52N$ contribute positively. The bounds are $10N$ up to $65N/2$ and $14N$ afterwards. The prime number theorem therefore gives

$$\log(J/I) \le (488 + o(1))N^2 = (61/288 + o(1))n^2.$$

This uniform bound, from [R08], is far larger than the available margin. Favorable finite rational evaluations receive much of their saving from large numerator primes. Those factors have not been controlled under the hypothetical equality $x = \theta$.

6.4 Direct output rounding gives a weaker sufficient target

There is no requirement that a successful correction have small coefficients: the arithmetic estimate is uniform over their size. We can therefore round directly in its b-dimensional output space. This removes some unnecessary geometric requirements of the earlier graph-lattice searches. The result combines [R06] and [R22].

Write the available coefficient lattice as $U = U_0 + Z\Gamma$, where Γ is an invertible integer b by b matrix. Set $\mathcal{L} = \Gamma W M(\theta)^{-1}$ and $F_0 = U_0 \Gamma^{-1}$. Then the relative determinant is $\det(I_b + \mathcal{L}(Z + F_0))$. Choose integer input vectors $z_1, \dots, z_b$ with independent images $y_j = \mathcal{L} z_j$, let p_j be their Euclidean Gram–Schmidt vectors, and put $d_j = \|p_j\|^2/4$.

In each affine column class $f + \mathbb{Z}^n$, round $-\mathcal{L}f$ to the lattice generated by the y_j, from the last orthogonal coordinate to the first. Round a fractional part t down with probability $1 - t$ and up with probability t. The output error has mean zero and covariance at most $\Sigma_0 = \frac{1}{4}\sum_j p_j p_j^{\mathsf{T}}$: conditional means vanish and variances are at most $1/4$, so the cross moments vanish in reverse rounding order. The input mean need not be zero, since the determinant sees only the output.

Round the columns independently, and let E be the output error matrix. If $e_k(d)$ denotes the kth elementary symmetric function, then

$$\mathbb{E} \det(I_b + E) = 1, \qquad \mathbb{E} \det(I_b + E)^2 \le \mathcal{H}(d) := \sum_{k=0}^{b} k! e_k(d). \tag{6.6}$$

To verify the second formula, expand into principal minors. Cross terms with different column sets vanish by independence and zero means. For a fixed k-set, multilinearity and the covariance bound give at most $k!$ times the corresponding principal minor of Σ_0. Summing gives (6.6).

Some admissible correction consequently has relative absolute value at most $\sqrt{\mathcal{H}(d)}$. If all admissible corrections are nonzero, as under rationality at the retained prime scales, this choice is nonzero. Without that premise, there is still a nonzero choice of size at most $\mathcal{H}(d)$: if a is the smallest nonzero absolute value among the finitely many outcomes Y, then $a \le a\mathbb{E}|Y| \le \mathbb{E}Y^2$ because $\mathbb{E}Y = 1$.

Since $\mathcal{H}(d) \le b! \prod_j (1 + d_j)$, a sufficient uniform condition is

$$\sum_{j=1}^{b} \log(1 + d_j) = o(n^2). \tag{6.7}$$

It preserves the whole leading canonical margin. This condition is weaker than bounding the largest direction and charging it b times. It has not been proved for any explicit family of directions here. For the 23 finite-difference bases tested at each of $n = 48, 96$, the best values of $\log \mathcal{H}(d)/(2n^2)$ were about $.367, .389$, respectively, against a margin about $.00001889$ on the same scale. For rounding in a different input metric, let v_i be the resulting Gram–Schmidt input vectors and define the covariance upper bound

$$\Sigma = \frac{1}{4} \sum_i (\mathcal{L}v_i)(\mathcal{L}v_i)^{\mathsf{T}}.$$

For the same ordered input basis, the triangular-minor argument in [R22] gives $e_k(\Sigma) \ge e_k(d)$. Thus changing the metric cannot improve this bound for the tested bases. The actual rounding covariance also depends on the fractional parts and need not equal Σ.

6.5 A polynomial left inverse gives an exact bound, but loses the target

Finite calculations found unexpectedly small-degree left inverses for the pencil on the kernel of W. The following result explains both what they control and why the tested way of using them cannot provide the missing decay. This is the final reduction in [R09].

Allow the more general coefficient lattice $U = U_0 + Z\Gamma$, with $\Gamma \in \mathbb{Z}^{b \times b}$ invertible. Set $k = n - b$, choose rational matrices H_0, K_0 with

$$WH_0 = 0, \quad WK_0 = I_b, \quad T = [H_0 \; K_0] \text{ invertible},$$

and put $D(X) = M(X)H_0$, $C(X) = M(X)K_0$. The stack satisfies

$$S(X)T = \begin{pmatrix} D(X) & C(X) + U_0 \\ 0 & \Gamma \end{pmatrix}, \qquad S(X) = \begin{pmatrix} M(X) + U_0 W \\ \Gamma W \end{pmatrix}. \tag{6.8}$$

This displays the k directions killed by the correction separately from the b directions it can change.

Suppose a polynomial matrix $E(X)$ of degree at most d_0 satisfies $E(X)D(X) = I_k$. Let $\eta(E) > 0$ be the rational content of all coefficients of all k by k minors of E. If $x = a/c$ is in lowest terms, Cauchy–Binet in $\det(E(x)D(x)) = 1$ gives

$$\frac{c^{d_0 k}}{\eta(E) g_k(D(x))} \in \mathbb{Z}_{>0},$$

where g_k denotes rational content of the k-minors. Selecting all b bottom rows in (6.8) shows that $|\det\Gamma| g_k(D(x))/(|\det T|\delta(x))$ is a positive integer. Multiplying these relations proves

$$\frac{C_E c^{d_0 k}}{\delta(x)} \in \mathbb{Z}_{>0}, \qquad \delta(x) \le C_E c^{d_0 k}, \qquad C_E = \frac{|\det\Gamma|}{|\det T|\eta(E)}. \tag{6.9}$$

Thus the degree alone is not the cost; the normalization C_E matters.

To apply this bound to an admissible correction class, we also require a positive integer multiplier a_N that clears every determinant polynomial in that class:

$$a_N P_{N,U_0+Z\Gamma}(X) \in \mathbb{Z}[X] \qquad \text{for every } Z \in \mathbb{Z}^{n\times b}.$$

Theorem 4.2 supplies this for its scalar class $U = U_N^0 + m_N Z$. Section 4 of [R08] supplies it for the enlarged admissible class, with the same leading arithmetic cost. Integrality and invertibility of Γ alone do not give this clearing property.

For either admissible class, there is a decisive obstruction to using (6.9). At $x = 0$, every cleared determinant is integral. Finite differences then show that a_N clears all mixed minors, so $a_N\delta(0)$ is a positive integer. Applying (6.9) at zero gives

$$a_N C_E \in \mathbb{Z}_{>0}. \tag{6.10}$$

Consequently, after multiplying the bound at a/c by its required clearing factor $c^n a_N$, the proposed upper bound is $(a_N C_E)c^{n+d_0 k} \ge 1$. It cannot certify a cleared value below one at even a single size. Optimizing the left inverse or its coefficient content does not change this conclusion.

The loss has an identifiable source. Selecting all bottom rows removes $C(X) + U_0$; replacing the remaining minors by coefficient content also removes their dependence on the particular analytic value θ. The resulting estimate holds at every rational x. It is a limitation of this estimate, not a lower bound for the error of an actual correction.

6.6 Small-degree kernel inverses do not bound correction loss

A concrete family shows why favorable kernel data are insufficient on their own. Let $D_e(X)$ have columns $e_j - Xe_{j+1}$, $0 \le j < e$, in $e+1$ rows. A left inverse has entries X^{i-j} for $j \le i < e$ and zero otherwise, including the last column. Its degree is $e-1$, its coefficients are zero or one, and a left null row is $r_e(X) = (X^e, \dots, X, 1)$.

Take $6N$ chains of length six and N chains of length five. This gives precisely $n = 48N$, $k = 41N$, $b = 7N$, with a degree-five left inverse. For coefficient degrees $d = 0, \dots, 4$, the left-multiplication map has rank $(d+1)48N$; at $d = 5$ its rank is $287N$. These are the same profiles found at the tested moment sizes. The rank formula follows because the kernel on one chain consists of $f_j = X^{e-j}h$, with $\deg h \le d - e$.

Set $s = 3^t$, $m = s + 2$. Complete each chain by a column $m^{-1}e_e$; the resulting square matrix has determinant m^{-b}. Take the correction row on each block to be $(1/2 - 1/m)e_e^{\mathsf{T}}$ and the initial correction column to be e_e. Block elimination gives

$$\det(M(X) + (U_0 + mZ)W) = 2^{-b}\det(I_b + sR_0(X)Z),$$

where R_0 consists of the block rows r_e. At an integer x, the matrices $R_0(x)Z$ range over all integer b by b matrices, since each r_e ends in one. The attained values are therefore $2^{-b}(1 + s\mathbb{Z})$, whose least absolute value is 2^{-b}. The loss relative to the canonical determinant is $(m/2)^b$. With $t = n$, its logarithm divided by n^2 tends to $(7/48)\log 3 > 0$.

At a fixed rational $x = a/c$, a clearing factor $2^b c^k$ has logarithm $O_x(n)$. For $t > 6v_3(c)$ the correction determinant is nonzero by its reduction at 3. Thus this example has simple kernel inverses, small arithmetic cost and nonvanishing, yet a large correction loss. It does not reproduce the special moment identities or their displacement relations; it shows why the kernel profile by itself cannot replace those missing analytic estimates.

6.7 What successful finite approximations do and do not imply

There is also a useful explanation for very small corrected values in finite numerical searches. Let $M(X) = A + XB$ with rational A, B, W, $\operatorname{rank} W = b$, and let H_0 span $\ker W$. Suppose $\operatorname{rank}[AH_0\ BH_0] = n$ and $M(\theta)$ is invertible. Then

$$\Gamma W M(\theta)^{-1}\mathbb{Z}^n \text{ is dense in } \mathbb{R}^b \quad \Longleftrightarrow \quad \theta \notin \mathbb{Q}. \tag{6.11}$$

For rational θ the image is contained in a rational lattice. For irrational θ, a failure of density would give a nonzero row y such that $v = y\Gamma W M(\theta)^{-1} \in \mathbb{Z}^n$. Then $v(AH_0 + \theta BH_0) = 0$. Rationality of these two matrices forces both terms to vanish separately, and their joint rank forces $v = 0$, hence $y = 0$, a contradiction. We used the elementary character criterion: a subgroup of $\mathbb{R}^b$ is dense precisely when no nonzero real linear functional takes integer values on it.

The joint-rank condition was checked at $n = 48, 96$ by nonzero minors modulo 1000003, with residues $306830, 229640$. The numerical nonsingularity at θ was checked with intervals. At these sizes, irrationality would already imply arbitrarily fine correction with unrestricted coefficients. This explains why a finite search can work very well without proving the needed uniform theorem. Bounds on the attainable determinants must still be strong enough to preserve the decay as N grows. Controlling coefficient sizes is one possible way to obtain such bounds; Theorem 4.2 itself permits corrections of any size.

7 Two direct constructions for small beta sets

7.1 Saving zeros by allowing more than one value

To prove that at least one of $\beta(4), \beta(6)$ is irrational, it would suffice to construct nonzero integer linear forms in these two values tending to zero. Under joint rationality, a single fixed denominator would make all these forms integers. There is no need to cancel either retained coefficient. The same applies to the three-value set $\{\beta(4), \beta(6), \beta(8)\}$.

The differentiated rational-function method, as used by Krisman [3, Sections 2–4, especially Lemma 4.6], provides a natural test. Differentiating twice shifts the first even period from $\beta(2)$ to $\beta(4)$. Its usual numerator zeros have multiplicity at least three, so they survive the differentiation. Could simple zeros save enough numerator degree to compensate for the rational boundary terms they leave? This is the question tested in [R29], following [R28].

Let n, r, v be positive odd integers, and let $A > rv$ be even. Define

$$R_n(t) = (2^n n!)^{A-rv} \frac{\displaystyle\prod_{j=0}^{rn-1} (t - (r-1)n + 2j + 1)^v}{\displaystyle\prod_{h=0}^{n} (t + 2h)^A}. \tag{7.1}$$

The factorial normalization cancels the $n \log n$ terms in Stirling estimates. It does not change the primitive integer form, except possibly its sign. Reflection gives $R_n(-2n-t) = -R_n(t)$. If

$$R_n(t) = \sum_{h=0}^{n} \sum_{i=1}^{A} \frac{a_{i,h}}{(t+2h)^i},$$

then $a_{i,n-h} = (-1)^{i+1} a_{i,h}$ and $\sum_h (-1)^h a_{i,h} = 0$ for odd i.

Put $u = (r-1)n/2$ and $S_K(s) = \sum_{j=0}^{K-1} (-1)^j/(2j+1)^s$. The full sum and its tail are

$$F_n = \sum_{j \geq 0} (-1)^j R_n''(2j+1)/2, \qquad T_n = \sum_{j \geq u} (-1)^j R_n''(2j+1)/2.$$

Partial fractions give

$$\begin{aligned} T_n &= q_u + \sum_{s=4,6,\ldots,A+2} \rho_s \beta(s), \\ \rho_{i+2} &= \binom{i+1}{2} \sum_h (-1)^h a_{i,h}, \qquad q_u = -\sum_{h,i} (-1)^h \binom{i+1}{2} a_{i,h} S_{u+h}(i+2). \end{aligned} \tag{7.2}$$

The constant sum includes *all* i, including odd i. Their aggregate beta coefficients vanish, but their rational finite sums need not. Setting $u = 0$ gives the full sum and q_0, so

$$q_0 - q_u = \sum_{j=0}^{u-1} (-1)^j R_n''(2j+1)/2. \tag{7.3}$$

For $v \geq 3$ this prefix is zero. For $v = 1$ it generally is not.

7.2 The zeros remove large denominator primes by grouping

The zeros serve an arithmetic purpose as well as making terms vanish. Suppose a proper rational function $V(t) = \sum_{h,i} c_{i,h}(t+2h)^{-i}$ has p-integral coefficients and vanishes at each negative odd integer from $-2n+1$ to -1, where $p > n$ is odd. In the rational constant of its alternating sum, group terms by that negative odd integer x. The group is

$$-\chi(x) \sum_{h : x+2h>0} \sum_i \frac{c_{i,h}}{(x+2h)^i}, \qquad \chi(x) = (-1)^{(x-1)/2}.$$

The sum over all h is $V(x) = 0$, so the complementary range $x + 2h < 0$ represents the same group with the opposite sign. A denominator divisible by p cannot occur in both ranges, since that would give $p \mid 2(h-h')$ with $0 < |h-h'| \leq n < p$. At least one representation is therefore p-integral.

For (7.1), the coefficients are p-integral for odd $p > n$: local expansion only divides by differences $2(k-h)$. The required zeros survive for $V = R_n''/2$ when $v \geq 3$, and for $V = WR_n$ without differentiation when W is an integer polynomial and the product is proper. These constructions have no odd denominator primes above n in their rational constants. The simple-zero derivative does not enjoy this cancellation.

7.3 An exact residue formula for the extra cost

Take $v = 1$ and an odd prime satisfying

$$\max(rn, A+2) < p \leq (r+1)n - 1, \qquad t = \frac{(r+1)n - 1 - p}{2}.$$

In the finite sums for q_u, the only positive denominator divisible by p is p itself. It occurs at poles $h = n - t + j$, $0 \le j \le t$. At such a pole exactly one numerator factor is $-p$. If K_h is the product of the other factors divided by the other pole factors, local expansion gives $a_{A,h}/p \equiv -K_h$ and $a_{A-1,h} \equiv K_h \pmod p$. After multiplying q_u by p^{A+1}, all lower orders vanish. The two surviving binomial coefficients differ by A, and direct factorial evaluation gives

$$p^{A+1}q_u \equiv \frac{A}{2}\chi(p)(-1)^{n-t}(n!)^{A-r}\sum_{j=0}^{t}\frac{j!(rn-1-j)!}{((n-t+j)!)^A((t-j)!)^A} \pmod p. \tag{7.4}$$

Every factorial here has argument below p. This proves $v_p(q_u) \ge -(A+1)$, with equality whenever the residue is nonzero. Since all beta coefficients are p-integral, equality forces p^{A+1} into the full primitive multiplier.

For $r \ge 3$, the full constant q_0 is p-integral because its finite denominators are at most $2n-1 < p$. Thus these primes are a new cost of removing the prefix. For example, at $n = 25, A = 4, r = 3$, each of $79, 83, 89, 97$ occurs to exponent five in the primitive tail multiplier and is absent from the full-sum cost. The nonvanishing qualification in (7.4) matters: among the 67 recorded comparisons, one residue vanishes. At $n = 9, A = 2, r = 1, p = 11$, the actual constant valuation is -2, rather than -3.

7.4 Keep the zeros and cancel $\beta(2)$ instead

A second test avoids differentiation. Form the full sums of R_n and $(t+n)^2R_n$, with $v = 1$. Both have periods through $\beta(A)$, starting at $\beta(2)$, and their constants have the grouped cancellation just proved. Multiplication at a pole gives

$$a_{i,h}^{(2)} = (n-2h)^2a_{i,h} + 2(n-2h)a_{i+1,h} + a_{i+2,h}.$$

Call the two complete forms U_0, U_2, and their $\beta(2)$ coefficients b_0, b_2. With $g_2 = \mathrm{cont}_{\mathbb{Q}}(b_0, b_2)$, the combination

$$H_n = (b_2/g_2)U_0 - (b_0/g_2)U_2$$

uses primitive integer weights and cancels $\beta(2)$ exactly. For $A = 6$ it retains $\beta(4), \beta(6)$; for $A = 8$ it retains the three-value set. Its complete coefficient vector is then reduced to primitive form once more. The weights depend on n, so the fixed multiplier theorem of Section 2 does not apply.

7.5 The finite comparisons

The differentiated tests use $n = 1, 3, 5, 9, 17, 25$. At $n = 25$, the natural logarithms of absolute values, divided by n, are as follows. The primitive columns include the rational constant and the complete coefficient gcd, as in Lemma 1.2.

Retained values	A	r	v	Raw tail	Primitive tail	Primitive full sum
$\beta(4)$	2	1	1	$-.06$	7.48	7.48
$\beta(4), \beta(6)$	4	1	1	$-.13$	11.28	11.28
$\beta(4), \beta(6)$	4	3	1	-2.85	18.15	13.37
$\beta(4), \beta(6)$	4	1	3	$-.18$	8.96	8.96
$\beta(4), \beta(6), \beta(8)$	6	3	1	-5.80	22.79	17.35
$\beta(4), \beta(6), \beta(8)$	6	5	1	-4.48	37.78	20.36
$\beta(4), \beta(6), \beta(8)$	6	1	3	$-.26$	10.87	10.87
$\beta(4), \ldots, \beta(12)$	10	3	3	-5.44	15.69	15.69

For $A = 4, r = 3$, removing the prefix saves about 3.81 in raw size but adds about 8.59 in constant-denominator cost, in this normalization. The resulting primitive form is larger. These are finite values, not claimed limiting exponents. The last row keeps five even values: with $r = v = 3$, the degree condition $A > rv$ requires $A \geq 10$. It shows the larger support needed to retain triple zeros on that longer numerator block.

The cancellation alternative gives the following primitive values.

Retained support	A	r	$\log \lvert\text{primitive } H_{25}\rvert/25$
$\beta(4)$	4	3	10.88
$\beta(4), \beta(6)$	6	1	12.39
$\beta(4), \beta(6)$	6	3	13.13
$\beta(4), \beta(6)$	6	5	20.53
$\beta(4), \beta(6), \beta(8)$	8	3	15.36
$\beta(4), \beta(6), \beta(8)$	8	5	22.03
$\beta(4), \beta(6), \beta(8)$	8	7	32.61

For $A = 6, r = 3$, the two primitive input forms have rates about 3.13 and 3.10. Eliminating $\beta(2)$ raises the result to 13.13; the primitive cancellation weights have up to 429 bits. The $A = 4$ control is the earlier adaptive-isolation family, up to scalar and sign.

All 115 final primitive forms at the tested $n \geq 3$ have absolute value greater than one. Some first-index forms are small: the $A = 6, r = 3, n = 1$ cancellation gives $731 - 182\beta(4) - 552\beta(6)$. Its isolated smallness gives no sequence tending to zero. The tests identify costs of these two changes; they do not rule out a different small-set construction. In particular, Krisman's shorter finite-set theorem uses a specifically balanced rational function and its arithmetic savings. Testing the simple families above does not test every choice in that method.

The checks reconstruct all partial fractions by a second algorithm, retain every constant term, and compare all 222 complete forms, including intermediate ones. Numerical signs and sizes are enclosed both through Arb Hurwitz zeta values and through an Euler-transformed alternating series. The latter has positive differences

$$\Delta^j f_s(0) = \frac{4^j (j!)^2}{(2j+1)!} h_{s-1}\left(1, \tfrac{1}{3}, \ldots, \tfrac{1}{2j+1}\right), \qquad f_s(k) = (2k+1)^{-s},$$

where h_d is the complete homogeneous symmetric polynomial. Summing $2^{-j-1}\Delta^j f_s(0)$ and bounding its remaining positive tail supplies an independent numerical enclosure.

8 Other tested variants

8.1 The three-shift digamma construction

Before the mixed moments, [R15] adapted Litsey's divided-difference construction and rational-antidifference argument [4, Sections 3–6]. It gives an all-slope nonvanishing result, although the arithmetic and real estimates did not close in the tests. Let $\psi = \Gamma'/\Gamma$ be the digamma function. For rational λ, put

$$F(z) = \psi(z + 1/4) + (24\lambda - 5/2)\psi(z + 1/2) + (3/2 - 24\lambda)\psi(z + 1).$$

The weights sum to zero, and $F'(0) = 8(G + \lambda\pi^2)$. Its increment

$$\Delta(z) = F(z+1) - F(z) = \frac{1}{z + 1/4} + \frac{24\lambda - 5/2}{z + 1/2} + \frac{3/2 - 24\lambda}{z + 1}$$

is rational. At nonnegative integer nodes define $f_t = \sum_{j=0}^{t-1} \Delta(j)$ and $d_t = \sum_{j=0}^{t-1} \Delta'(j)$. Choose a row-node set R and a column-node set C, each of size $N = r+m$, whose union is $\{0, \dots, r+2m-1\}$ and whose intersection is a consecutive block of $r \geq 1$ nodes. The matrix entries, for $t \in R$ and $u \in C$, are

$$M(t,u;X) = \frac{f_u - f_t}{u-t} \quad (t \neq u), \qquad M(t,t;X) = X + d_t.$$

At $X = F'(0) = 8(G + \lambda\pi^2)$, these entries are the divided differences of F, with $F'(t)$ on a common-node diagonal. The determinant is a rational polynomial in this single target. The following proposition supplies nonvanishing for the determinant together with its common-node cofactors.

Proposition 8.1. *For every complex X, this square matrix has corank at most one. If it is singular, some common-node diagonal cofactor is nonzero. Thus the determinant and these cofactors have no common complex zero.*

Proof. Let $Q_C(z) = \prod_{u \in C}(z-u)$. A right kernel vector corresponds to polynomials U, W of degree below N through

$$U = Q_C \sum_{u \in C} \frac{v_u}{z-u}, \qquad W = Q_C \sum_{u \in C} \frac{v_u f_u}{z-u}.$$

The kernel equations are precisely $W(t) = f_t U(t)$ at every union node and $W'(t) = f_t U'(t) + (X + d_t)U(t)$ at each common node. The transpose has the same interpolation conditions.

For two such pairs, $W_1U_2 - W_2U_1$ has $2N$ zeros counted with multiplicity and degree at most $2N-2$, so it vanishes. All pairs therefore have the same reduced ratio. If the kernel dimension is d, canceling the highest $d-1$ coefficients of their polynomial multipliers gives a pair of maximum degree at most $N-d$. Write $\Delta = n_\Delta/d_\Delta$ in lowest terms, with denominator degree $q \leq 3$. The zero sum of the weights gives $\deg n_\Delta \leq q-2$. For a pair of maximum degree D, form

$$T(z) = d_\Delta(z)\{W(z+1)U(z) - W(z)U(z+1)\} - n_\Delta(z)U(z)U(z+1).$$

Its degree is at most $2D+q-2$. It vanishes at every adjacent pair of union nodes and has a double zero at each adjacent pair of common nodes, giving $2N-2$ zeros. If $d \geq 2$, its degree is at most $2N-3$, so $T = 0$. This would make W/U a rational antidifference of Δ. But a rational difference has residue sum zero in each pole class modulo $\mathbb{Z}$, whereas the class of $-1/4$ contains one pole of Δ with residue one. Hence $d \leq 1$.

If U vanished at all common nodes, so would W. Dividing by their common-node product leaves degrees at most $m-1$ and preserves the value conditions at all union nodes. If $m = 0$, the reduced pair is already zero. For $m \geq 1$, the analogous polynomial T has degree at most $2m-1$ but at least $r+2m-1 \geq 2m$ zeros, again impossible. Some common node therefore has $U(t) \neq 0$. Both the left and right kernel coordinates there are nonzero, so the corresponding diagonal adjugate entry is nonzero. □

For $\lambda = 1$ and border ratio $2m/r = 1/5$, put $P(X) = \det M(X)$ and $\xi = 8(G+\pi^2)$. Let $K = 1/\operatorname{cont}_{\mathbb{Q}}(P) > 0$, so KP is the complete primitive integer polynomial. The following values are rounded from certified finite enclosures; every logarithm is divided by r^2:

r	Multiplier cost $\log K/r^2$	Raw decay $-\log\lvert P(\xi)\rvert/r^2$	Primitive growth $\log\lvert KP(\xi)\rvert/r^2$
20	3.384060	3.168234	.215826
40	3.675448	3.417346	.258102
80	3.771926	3.541927	.229999
160	3.813989	3.604388	.209601

The last column is the first minus the second. Its positive entries mean that the primitive values have absolute value greater than one at these sizes, after including the full coefficient gcd. The slope $\lambda = 1/16$ is a warning against extrapolation: its primitive rates are negative at $r = 20, 40, 80$, but positive at $r = 160$. Litsey's special slopes serve as controls; their arithmetic cancellation cannot be assumed at every rational slope.

8.2 Two further Padé variants

The reassessment [R28] tested two changes to the family of Section 3. First, writing $X = t(t-1)$ converts powers X^j into the even factorial basis through

$$X^j = \sum_{\ell} S(j,\ell) T_{2\ell}(t), \qquad S(j+1,\ell) = S(j,\ell-1) + \ell(\ell+1) S(j,\ell).$$

The first nontrivial Hankel determinant of the resulting remainders is $E_0(E_2 + 2E_1) - E_1^2$. Nine tests constructed the entire polynomial in $\beta(4)$ and reduced it to primitive integer coefficients. At seed index $n = 12$, determinant sizes $1, 2, 4, 6$ gave logarithms of absolute primitive values approximately $78.77, 164.19, 324.32, 508.39$. The coefficient calculations are exact; these decimal evaluations were exploratory. No estimate uniform in growing determinant size was obtained.

Second, lowering the numerator multiplicity and retaining the crossed rational residues gives the real integral

$$J_{n,m} = \frac{2}{(n!)^{m-2}} \int_0^\infty \frac{y\mu'''(y) \prod_{j=1}^{n/2} (y^2 + (j-1/2)^2)^m}{\prod_{j=0}^{n} (y^2 + j^2)} \, \mathrm{d}y, \qquad \mu(y) = \frac{1}{2\cosh \pi y}.$$

Partial fractions express it in $\mathbb{Q} + \mathbb{Q}\beta(4)$. The $m = 5$ case agrees exactly with the earlier construction at the four comparison indices. For $m = 3, n = 2$, its primitive form is $90\beta(4) - 89$, of absolute value about .00501. At $n = 4, 8, 12, 20$, the corresponding logarithms are approximately $12.50, 37.93, 61.43, 121.56$. Lower multiplicity reduces a cost, but these tests supply no favorable asymptotic comparison.

A Catalog of the 29 research records

The catalog below groups earlier intermediate estimates under their strongest final version. An entry described as a finite test is not an all-size obstruction. Record identifiers also index the companion coverage manifest, which retains the source proof hashes and identifies the selected evidence. The complete private research archive is not required to build this manuscript or run its packaged checks.

R01. Uniform arithmetic The common congruence class gives the coefficient cost approximately 1.870614. The proof includes the mixed 2-adic factorization, odd-prime rank savings and the Chinese remainder construction.

R02. Boundary borders and recurrence lift The mixed recurrence has two additional boundary moment rows. Coupled first-order integer recurrences generate them from two rational seeds. A larger determinant represents the same forms with scalar cost $O(n \log n)$; its primitive polynomial is unchanged. Two natural borders at $n = 24, 48$ remain unfavorable. The full matching correction is not rank two at those sizes.

R03. Content of the combined row lattice The attainable progression is normalized by the rational content of the stacked maximal minors. Its remaining integer step has logarithm $O_x(n \log n)$, so the content determines the leading rate at a fixed rational evaluation.

R04. Prime-scale nonvanishing and covering growth Three seed determinant polynomials have no rational root, yielding nonvanishing for every fixed rational slope at all sufficiently large prime scales. The same record derives the correction-loss criterion; finite covering growth does not prove it uniformly.

R05. Dual obstruction to bounded approximation A graph-lattice covering argument certifies the finite approximation for every affine class at the tested sizes, within a prescribed coefficient cap. Exact dual equations show why unrestricted density cannot be assumed in an irrationality proof. The required bounded, uniform covering estimate remains open.

R06. Determinant averaging in the enlarged lattice Unbiased rounding controls the determinant by a covariance second moment, charging the error directions jointly. This weakens the earlier maximum-norm target. The calculations at $n = 48, 96$ do not prove the resulting bound as n grows.

R07. Row and column Gram energies Gram Cauchy–Schwarz separates row and column volumes. A uniform column-energy inequality is proved, and the deterministic endpoint repair is retained in the row Gram. Later records sharpen both quantitative estimates.

R08. Index comparison and a larger coefficient lattice The uniform index-loss upper bound is $61/288$ on the n^2 scale, much larger than the available margin. Removing unnecessary coefficient congruences enlarges each row lattice by an index factor $\exp((5/48 + o(1))n^2)$, while retaining the proved arithmetic cost and prime-scale nonvanishing. That extra freedom has no proved uniform real estimate. All 393 tested full-variation-row minors also miss the required rate.

R09. Polynomial kernel inverses A normalized left-inverse bound controls determinant content but necessarily costs at least the arithmetic threshold. A comparison family has the same dimensions and degree-five profile with a positive quadratic correction loss. Conditional density explains finite successes without establishing irrationality.

R10. Low-degree correction freedom Corrections below the column band preserve the target period and endpoint zero. A local Smith-image optimization attains the minimum two-layer denominator cost. The first global choices lose other arithmetic and real savings; they are superseded by the uniform class.

R11. Matching the mixed slope Changing D to $D - 16\lambda E$ retains $G + \lambda\pi^2$. Linear high-degree completions satisfying the same band and endpoint conditions give exactly the same determinant. Full, band and endpoint completions were compared with complete primitive normalization.

R12. Moment recurrences and natural bases The Chebyshev recurrence acquires lower and upper band defects after period matching. Their forced central dyadic defect cannot be removed by the specified integral corrections; a separate rational obstruction rules out a homogeneous extension. Natural recurrence bases, even after removing column content, fail the finite covering criterion.

R13. Moment-derived directions An interior displacement formula has three boundary terms. Saturated moment-derived directions still fail the finite determinant test. A proposed scalar kernel-pencil route retains a rational approximation error. Degree-five polynomial left inverses of the full low-moment kernel are certified at $n = 48, 96$; no all-size existence or decay theorem is inferred.

R14. Limit of the absolute-value interpolation bound Dual probability measures prove that the absolute-value potential method has integrated cost at least 0.04086151228157811175. With the then-current column and repair bounds, it needed to be below 0.037728119494102. This limits that estimate, not the actual row determinant. Sharpening the repair and column bounds later bypasses the limitation.

R15. Three-shift digamma determinants The determinant and common-node cofactors have no common complex zero for every rational slope. Exact full primitive polynomials still have unfavorable values for the generic tested slopes. Alternative borders, nonzero-sum weights and two four-shift families were exploratory tests, not asymptotic certificates.

R16. The rational image of integer corrections Smith normal form and an explicit last-row construction show that unrestricted correction determinants attain every value in one arithmetic progression. Mixed-minor content gives its step; the result imposes no height bound on an attaining correction.

R17. Real lattice searches in the paired class Integer lattice searches within the protected paired-row congruence class give nonzero primitive values below one at $n = 24, 48, 96$. Exact polynomial normalization and interval evaluation certify each example. They do not supply an unbounded irrationality sequence.

R18. The real row operator The polynomial-part projection is a contraction in the relevant weighted norms. Its failure to commute with the endpoint factor is an explicit finite-rank moment operator. This supplies the main-row plus repair decomposition used in the final estimate.

R19. Repair Gram minors and positive canonical margin Compound matrices, factorial Schur comparison and a damped energy bound give repair cost below 0.00498. A complete degree-200 column certificate yields a positive canonical margin. The audit covers the stated dependencies, but not later uncontrolled integer corrections.

R20. Spectrum of the main-row operator Paired rows isolate an explicit Jacobi volume. Rational approximation of the remaining operator bounds its singular values. Twenty-six global potential certificates give spectral cost 0.0450807972 and the main-row rate used in the final estimate.

R21. Paired corrections and protection at two Copying paired correction rows preserves the original fixed rational rows. Explicit dyadic row divisibility protects the exact determinant valuation at two while allowing most odd-prime savings. The first protected global choices still worsen real decay; later lattice searches address that finite defect.

R22. Structured directions and direct output rounding Only b independent integer image directions are needed for rounding. The covariance-moment bound for a fixed ordered sequence of image subspaces is optimal over rounding metrics. Twenty-three finite-difference bases at each of $n = 48, 96$ miss the required margin.

R23. First uniform endpoint-repair bound Rank-one truncations control every singular value of the repair and give cost at most $0.009486012 + 0.178211892/n$. This is superseded numerically by the Gram-minor bound, while its kernel identity and projection argument remain in use.

R24. Finite real approximation in the uniform class The new all-row scalar-modulus class, including divisibility by 2^C, admits certified small nonzero primitive values at $n = 48, 96$. These tests use the exact uniform arithmetic family, unlike the earlier paired searches; a uniform real bound remains open.

R25. Individual $\beta(4)$ isolation on an odd grid Fixed polynomial multipliers cannot isolate $\beta(4)$ in the stated valuation-separated class. Adaptive finite multipliers do isolate it but have unfavorable primitive values. Explicit telescoping multipliers give zero forms, and a finite quotient calculation is kept distinct from an all-degree classification.

R26. Real interpretation of the Padé family The odd-m Kawashima–Poëls construction gives separate $\beta(4)$ remainders without eliminating $\beta(2)$. A real Cauchy representation, moving half-integer evaluation and complete rational tail conversion give the stated forms and family nonvanishing.

R27. Padé remainder rates and arithmetic cost A signed saddle calculation gives the actual remainder rate, including eventual nonvanishing. An explicit improved clearing factor still exceeds that decay for every fixed evaluation ratio at least one half. Further primitive coefficient content is not excluded.

R28. Reassessment and changes of target The useful relaxations are a small set of unresolved even values and rational relations among $1, \pi^2, G$. Simple KP Hankel determinants and lower seed multiplicity were tested with complete coefficients; neither gave favorable growing examples. Other target changes were assessed without claiming new results.

R29. Small-set tails and $\beta(2)$ elimination Second derivatives with simple zeros can add large-prime constant costs; triple zeros retain grouped cancellation. Undifferentiated forms avoid those primes but their tested $\beta(2)$-elimination weights are costly. All 115 final forms at the tested indices $n \geq 3$ have primitive absolute value above one.

B Reproduction and audit scope

The companion directory contains the LaTeX source, a build command, rational trial data, certificate checkers and the coverage manifest. The check runner copies inputs into a fresh working directory, so replaying a certificate does not overwrite its recorded result. Its featured checks are the 26 row-potential bounds, the repair integral, the degree-200 column certificate and root counts, the three nonvanishing seed polynomials, and the complete small-set forms. Additional finite checks cover the elementary determinant reductions.

The independent mathematical audit of 8 November 2026 reported no gap affecting the canonical estimate and its specified arithmetic and nonvanishing inputs. Its independently written arithmetic reproduced the positive margin and checked 878800 moment congruences, local corrections in 70 prime cases, 313 seed determinant evaluations and 2880 Frobenius identities. It also checked the general repair argument, the uniform limiting passages, the main-row estimate and the complete column critical-point certificate. A public scope summary and the numerical outcome accompany this supplement. The audit accepted the stated classical results, including the factorial Schur tableau identity, Stirling's formula and the prime number theorem. It did not use the original Catalan irrationality claim or its published decay constant.

A further independent audit of the 41-page assembly on 10 November 2026 checked the isolation and prime-dilation arguments, the real Padé integral and rate, the determinant progression and rounding arguments, the small-set residue formula, and digamma nonvanishing. It identified one missing hypothesis in the left-inverse obstruction: the clearing multiplier must work throughout the correction class. The arithmetic results cited in Section 6.5 establish this hypothesis for both admissible classes. The canonical estimate survived this audit. Its mixed-moment arithmetic, nonvanishing and Gram-bound dependencies were checked against the pinned 8 November audit.

All 13 packaged checks passed in the auditor's scratch copy. Fresh calculations checked 240 Padé clearing cases and 18 product integrals, both degree-25 small-set tables, and the stated prime-valuation examples. The larger digamma table was compared with its saved enclosures; the remaining historical experiments were outside this replay. The audit accepted the cited Kawashima–Poëls theorems and classical factorial Schur identity as inputs. Its scope and the subsequent corrections are recorded separately in the companion files.

Neither audit nor finite replay supplies the uniform correction estimate (5.12) or (6.7).

The work was prepared with OpenAI Codex. The supplied calculations and the independent mathematical audits are not proof-assistant verification.

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